

Quicreate Skyblock RFA

Walkthrough · pack version 0.9.0 · Minecraft 1.20.1 Forge



A step-by-step route from a tree in the ocean to creative-tier Create parts on five worlds. Every recipe, drop chance and ore in this book was read from the pack's own files when it was built.

Built 2026-11-10 from the pack files · 220 quest steps · 11 chapters

How to use this guide

The guide follows the in-game quest book (Heracles) chapter by chapter, in the order you will play it. Each quest is one **step**, numbered **#1** onwards. A step shows what the quest asks for, the quest's own hint, the recipe (with the machine it needs), and for every ingredient *where it comes from*: an earlier step, a sifting table, an ore, a mob drop, fishing, or a short recipe.

Recipes shown are the ones that exist in this pack after the CraftTweaker scripts and the Quicreate datapack have changed them, so they can differ from the wiki. When an item has several recipes the guide shows the one that fits the route best and says how many others JEI will show you.

Worlds have a **what is here** box at the start of their chapter: terrain, ores (with heights), mobs and their drops, and the structures with the mobs and chest loot inside them. The appendices at the back hold the full sifting tables, the Mechanical Extruder list, fishing, card packs and spawn eggs, boss drops, and a list of anything the builder could not find a source for.



- Small number: how many. %: chance. #: any item of that tag. Blue number: millibuckets of fluid.
- **#12** links point to the step where that item is made.

The route

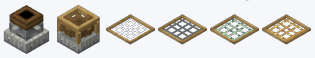
Island → Sifting → New Age (solar heat, brass, electricity) → Improvements (circuits, nuclear heat, steel) → Space (your first rocket) → the Moon (stronghold and the Ender Dragon: the soft ending) → Mars → Venus and Mercury → Glacio → the creative parts (the true ending). The Cards & Mobs side chapter can be done any time after you have a sifter.



The Island · Chapter 1: The Sky · 17 steps
Wood, cobblestone, andesite alloy and your first Create parts.

p. 4



Sifting · 1.1: Sifting · 25 steps
Mill cobblestone down to gravel, sand and dust, then sift them for metals, redstone and more.

p. 9



New Age · 1.2: New Age · 35 steps
Sheets and wires, solar heat instead of blaze burners, brass, and your first electricity.

p. 21



Improvements · 1.3: Improvements · 21 steps
Circuits, motors, nuclear heat and steel.

p. 35



Space · 1.4: Space · 18 steps
Steel plates, a space suit, rocket fuel and your first rocket.

p. 44



The Moon · Chapter 2: The Moon · 29 steps
Desh, diamonds and amethyst, the only stronghold, and the Ender Dragon (soft ending).

p. 52



Mars · Chapter 3: Mars · 19 steps
Ostrum, netherite, the Cursed Pyramid and the Ancient Remnant.

p. 63



Venus and Mercury · Chapter 4: Venus & Mercury · 20 steps
Calorite, the Harbinger and Umvuthi on Venus; platinum, Ignis and the Netherite Monstrosity on Mercury.

p. 73



Glacio · Chapter 5: Glacio · 16 steps
The last world, four structures and the materials for the creative parts.

p. 82



The Creative Ending · Endgame: Creative · 13 steps
Creative alloy, sheet, casing, motor and blaze cake: the true ending.

p. 88



Side chapter: Cards and Mobs · Side: Cards & Mobs · 7 steps
Card packs from sifting, spawn eggs from cards, the Wither and the Flight Anchor. Any time after the sifter.

p. 94

CHAPTER 1: THE SKY · STEPS #1–#17

The Island

You start on a small island in an endless ocean. The island template in `config/easy_skyblock_island/islands/default.nbt` is grass and dirt with one oak tree, and a chest holding a single Lava Bucket . Keep that lava: with water it is your first cobblestone generator.

Tips

- Your inventory is kept on death (see **#4 About This Modpack**).
- Vein mining (the `~` key, Ore Excavation) takes whole trees and their leaves at once.
- Andesite alloy in this pack is made with charcoal, so get a furnace and burn logs early.

#1 Logs

You can vein mine whole trees and leaves with ~

- Collect 16× **Any logs**

#2 Saplings

You can grow saplings quick by sneaking!

After **#1 Logs**

- Collect 4× **Any saplings**

#3 Get Crafting Table

After **#1 Logs**

- Collect 1× **Crafting Table**

Crafting Table vanilla



Ingredient	Where it comes from
any Planks	Crafting Table (shapeless): any Oak Log → 4



#4 About This Modpack

Click for details and motivation!

- **Read this** (tick it in the quest book)

> **This is a Skyblock Modpack based around rapid expandable progression.**
>
> I wanted a skyblock modpack you could sit down, feel progression in, and "complete" your first time in around 10 hours without a plan. Then after that, optimize your pathing and get faster on future playthroughs.
>
> The intial end goal is to beat the Ender Dragon of course, but strongholds only spawn on the moon! You will need to build a Rocket and make your way there to win the game.
>
> Also on the moon is an optional boss and new materials which can be used to access new technology, rockets, and weapons. You'll be able to travel to more distant planets, finding new bosses to defeat and more materials to improve your factory. The defeat of the Ender Dragon is the soft ending, but the true ending is access to the **Creative** factory components which will require defeating the bosses of every planet and a robust automated factory.
>
> Inventory is kept on death by default because I think it's a artificial difficulty in skyblock. Maybe you think it's not, this can be changed in the config.



#5 Ad Astra

Core Mod 1: Space

After **#4 About This Modpack**

- **Read this** (tick it in the quest book)

> Ad Astra is the modern space travel mod, comparable to Galacticraft or whatever was in tekkit.
>
> I wanted to try this mod the moment I found it, but it had the issue every space mod has - every planet is mostly empty, and you have no reason to engage with it beyond the intrinsic fun of making a rocket.
>
> This is a core mod because I wanted to make it relevant to late game progression to make it feel worth using, rather than adding it as a kitchen sink style addition. Planets have currently been improved somewhat with some generation and structures, and making the resources found on each planet critical to tech advancement.



#6 Create: New Age

Core Mod 2: Electricity

After **#4 About This Modpack**

- **Read this** (tick it in the quest book)

> Create: New Age was the first mod I found and I used it on its own in a regular world and liked it a lot! I made it a core mod because I think the physical nature of the power generation, and heat generation, this mod adds novel and unique.



#7 Cataclysm

Core Mod 3: Boss Fights

After **#4 About This Modpack**

- **Read this** (tick it in the quest book)

> Cataclysm adds great boss fights and more materials to add more angles of progression. The bosses are scattered across several planets to let you make tech advancements with combat rather than just exploration and mining.



#8 Cobblestone

Vein mining could also speed up cobble collection

After **#3 Get Crafting Table**

- Collect 32× **Cobblestone**

Item	Where it comes from
Cobblestone	Mechanical Extruder with Cobblestone below (Lava + Water) Cobblestone generator, lava flowing next to water (the starting chest has a lava bucket). (game mechanic)

#9 furnace

After #8 Cobblestone

- Collect 1× Furnace

Crafting Table vanilla



Ingredient	Where it comes from
any Cobblestone	Made in #8 Cobblestone Mechanical Extruder with Cobblestone below (Lava + Water) Cobblestone generator, lava flowing next to water (the starting chest has a lava bucket). (game mechanic)

#10 andesite

Crafted from 4 Cobblestone

After #8 Cobblestone

- Collect 8× Andesite

Item	Where it comes from
Andesite	Mechanical Extruder with Block of Andesite Alloy below (Lava + Water)

Or make it:

Crafting Table (shapeless) vanilla



+4 other recipes in JEI

Ingredient	Where it comes from
Diorite	Mechanical Extruder with Block of Quartz below (Lava + Water) Crafting Table: 2× Cobblestone + 2× Nether Quartz → 2 Chest loot in the Monastery (Mars), 44% per chest
Cobblestone	Made in #8 Cobblestone Mechanical Extruder with Cobblestone below (Lava + Water) Cobblestone generator, lava flowing next to water (the starting chest has a lava bucket). (game mechanic)

#11 Quicreate Items

Quicreate adds some new recipes and items to connect mods together

After #4 About This Modpack

- **Read this** (tick it in the quest book)

#12 Quicreate Weapons and Items

Adds some minor weapons for Boss fights

After #4 About This Modpack

- **Read this** (tick it in the quest book)

> Quicreate is the custom mod for this pack that adds a lot of recipes, resource compatability, new items and weapons using the core mods, and the custom structure and terrain generation changes for each planet!

#13 Quicreate Endgame

Add creative resources accessible near the end

After #4 About This Modpack

- **Read this** (tick it in the quest book)

> Create: Cataclysm gives us the the end game goal. After total planetary travel is achieved and every boss slain, you can craft the Creative Tier items for maximum factory convenience and potential.

#14 Charcoal

After #9 furnace

- Collect 8× **Charcoal**

Furnace vanilla



+1 other recipe in JEI

Ingredient	Where it comes from
any Oak Log	Starting island: 6× Oak Log

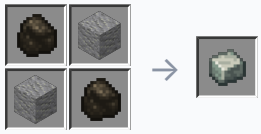
#15 Andesite Alloy

Charcoal and Andesite

After #14 Charcoal, #10 andesite

- Collect 16× **Andesite Alloy**

Crafting Table Quicreate



Crafting Table create



+5 other recipes in JEI

Ingredient	Where it comes from
Charcoal	Made in #14 Charcoal
Andesite	Made in #10 andesite Mechanical Extruder with Block of Andesite Alloy below (Lava + Water)

#16 Andesite Casing

After #15 Andesite Alloy

- Collect 4× **Andesite Casing**

Use item on block (hand or Deployer) create



Use item on block (hand or Deployer) create



Ingredient	Where it comes from
any Stripped Oak Log	Strip a log or wood block by right-clicking it with an axe. (game mechanic)
Andesite Alloy	Made in #15 Andesite Alloy

#17 Shafts & Cogwheels

The bones of every Create machine

After **#15 Andesite Alloy**

- Collect 8× **Shaft**
- Collect 4× **Cogwheel**

> Shafts carry rotation; cogwheels change its direction and speed.
> Cogwheels are a shaft plus planks.

Crafting Table create



Mechanical Saw create



+1 other recipe in JEI

Ingredient	Where it comes from
Andesite Alloy	Made in **#15 Andesite Alloy**

Crafting Table (shapeless) create



Deployer create



+1 other recipe in JEI

Ingredient	Where it comes from
Shaft	Crafting Table: 2× Andesite Alloy → 8
any Planks	Crafting Table (shapeless): any Oak Log → 4



1.1: SIFTING · STEPS #18–#42

Sifting

Sifting is how you get almost every raw material on the island. Mill cobblestone into gravel, gravel into sand and sand into dust, then put them through a Sifter with a mesh. Better meshes give better drops, and a **waterlogged** sifter has its own drop table. The full tables are in Appendix A.

Tips

- Meshes go in the sifter; the block you sift is the input. String mesh → andesite → zinc → brass → advanced brass.
- Card packs drop from many sifting tables at low chances. See the Cards & Mobs chapter.



#18 Hand Crank

Your first rotation

After **#16 Andesite Casing**, **#17 Shafts & Cogwheels**

- Collect 1× **Hand Crank**

Crafting Table create



Ingredient	Where it comes from
any Planks	Crafting Table (shapeless): any Oak Log → 4
Andesite Alloy	Made in **#15 Andesite Alloy**



#19 Water Wheel

Free rotation from flowing water

After **#18 Hand Crank**

- Collect 1× **Water Wheel**

Crafting Table create



Ingredient	Where it comes from
any Planks	Crafting Table (shapeless): any Oak Log → 4
Shaft	Made in **#17 Shafts & Cogwheels**

#20 Mill

After #18 Hand Crank

- Collect 1× **Millstone**

Crafting Table create



Ingredient	Where it comes from
Cogwheel	Made in #17 Shafts & Cogwheels
Andesite Casing	Made in #16 Andesite Casing
any Andesite	Made in #10 andesite Mechanical Extruder with Block of Andesite Alloy below (Lava + Water)

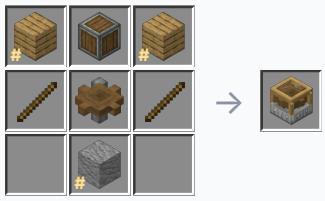
#21 Sifter

Different sifted blocks give different resources!

After #20 Mill

- Collect 1× **Sifter**

Crafting Table createsifter



Ingredient	Where it comes from
any Planks	Crafting Table (shapeless): any Oak Log → 4
Andesite Casing	Made in #16 Andesite Casing
Stick	Starting island: 53× Oak Leaves (drops Stick) Sift Dust block with String Mesh: **25%** Sift Dust block with Andesite Mesh: **15%** Crafting Table: 2× any Planks → 4
Cogwheel	Made in #17 Shafts & Cogwheels
any Andesite	Made in #10 andesite Mechanical Extruder with Block of Andesite Alloy below (Lava + Water)

#22 Gravel

Gravel comes from milled Cobblestone!

After **#20 Mill**

- Collect 16× **Gravel**

Millstone create



Crafting Table useful-flint



+1 other recipe in JEI

Ingredient	Where it comes from
Cobblestone	Made in **#8 Cobblestone** Mechanical Extruder with Cobblestone below (Lava + Water) Cobblestone generator, lava flowing next to water (the starting chest has a lava bucket). (game mechanic)

#23 String

Mill a sapling to get String

After **#20 Mill**

- Collect 8× **String**

Item	Where it comes from
String	Open a Light Kelpy Ball (fishing): 26.2% per ball Chest loot in the Ruined Citadel (The Moon), 57.8% per chest Chest loot in the Mars Temple (Mars), 22.9% per chest Chest loot in the Cursed Pyramid (Mars), 59% per chest Chest loot in the Desert Site (Mars), 59% per chest Chest loot in the Frosted Prison (Glacio), 57.8% per chest

Or make it:

Millstone Quicreate



+6 other recipes in JEI

Ingredient	Where it comes from
any Sapling	Covered later in **#26 New Saplings** Sift Dirt with String Mesh (waterlogged): **10%**

#24 Mesh

Different sifted blocks give different resources!

After **#21 Sifter**, **#23 String**

- Collect 1× **String Mesh**

> String mesh on dirt gives seeds; on sand, iron nuggets; on dust, gunpowder and sticks.

Crafting Table pack script



Ingredient	Where it comes from
Stick	Starting island: 53× Oak Leaves (drops Stick) Sift Dust block with String Mesh: **25%** Sift Dust block with Andesite Mesh: **15%** Crafting Table: 2× any Planks → 4
String	Made in **#23 String** Open a Light Kelpy Ball (fishing): 26.2% per ball Chest loot in the Ruined Citadel (The Moon), 57.8% per chest

#25 Iron

Sift sand with a string or andesite mesh

After **#24 Mesh**

- Collect 27× **Iron Nugget**

Item	Where it comes from
Iron Nugget	Sift Sand with Andesite Mesh: **50%** Sift Sand with String Mesh: **25%** Open a Light Kelpy Ball (fishing): 21.3% per ball Chest loot in the Lunar Tower (The Moon), 38% per chest Chest loot in the Lunarian Village (The Moon), 38% per chest Chest loot in the Sunken City (Glacio), 96.8% per chest

Or make it:

Encased Fan + Water (washing) create



+11 other recipes in JEI

Ingredient	Where it comes from
Crushed Raw Iron	Sift Sand with Brass Mesh: **30%** Sift Sand with Zinc Mesh: **10%**

#26 New Saplings

Waterlogged Sifters have different drops!

After **#24 Mesh**

- Collect 1× **Dark Oak Sapling**
- Collect 1× **Acacia Sapling**
- Collect 1× **Jungle Sapling**
- Collect 1× **Birch Sapling**
- Collect 1× **Spruce Sapling**
- Collect 1× **Bamboo**

> Did you know you can fill bottles with a water source, to fill a cauldron to make a second source? Faster than waiting for rain!

Item	Where it comes from
Dark Oak Sapling	Sift Dirt with String Mesh (waterlogged): **10%**

Or make it:

Crafting Table (shapeless) sapling-tweaks



+1 other recipe in JEI

Ingredient	Where it comes from
Bone Meal	Millstone: Koboleton Bone → 4 Lionfish drops it (5% per kill; where it spawns is not in the data files) Raw Cod drops it (5% per kill; where it spawns is not in the data files) Pufferfish drops it (5% per kill; where it spawns is not in the data files)

Item	Where it comes from
Acacia Sapling	Sift Dirt with String Mesh (waterlogged): **10%**

Or make it:

Crafting Table (shapeless) sapling-tweaks



+1 other recipe in JEI

Ingredient	Where it comes from
Bone Meal	Millstone: Koboleton Bone → 4 Lionfish drops it (5% per kill; where it spawns is not in the data files) Raw Cod drops it (5% per kill; where it spawns is not in the data files) Pufferfish drops it (5% per kill; where it spawns is not in the data files)

Item	Where it comes from
Jungle Sapling	Sift Dirt with String Mesh (waterlogged): **10%**

Or make it:

Crafting Table (shapeless) sapling-tweaks



+1 other recipe in JEI

Ingredient	Where it comes from
Bone Meal	Millstone: Koboleton Bone → 4 Lionfish drops it (5% per kill; where it spawns is not in the data files) Raw Cod drops it (5% per kill; where it spawns is not in the data files) Pufferfish drops it (5% per kill; where it spawns is not in the data files)

Item	Where it comes from

Birch Sapling	Sift Dirt with String Mesh (waterlogged): **10%**

Or make it:

Crafting Table (shapeless) sapling-tweaks



+1 other recipe in JEI

Ingredient	Where it comes from
Bone Meal	Millstone: Koboleton Bone → 4 Lionfish drops it (5% per kill; where it spawns is not in the data files) Raw Cod drops it (5% per kill; where it spawns is not in the data files) Pufferfish drops it (5% per kill; where it spawns is not in the data files)

Item	Where it comes from
Spruce Sapling	Sift Dirt with String Mesh (waterlogged): **10%**

Or make it:

Crafting Table (shapeless) sapling-tweaks



+1 other recipe in JEI

Ingredient	Where it comes from
Bone Meal	Millstone: Koboleton Bone → 4 Lionfish drops it (5% per kill; where it spawns is not in the data files) Raw Cod drops it (5% per kill; where it spawns is not in the data files) Pufferfish drops it (5% per kill; where it spawns is not in the data files)

Item	Where it comes from
Bamboo	Sift Dirt with String Mesh (waterlogged): **10%** Chest loot in the Sunken City (Glacio), 14.5% per chest Panda drops it (100% per kill; where it spawns is not in the data files)

#27 Sand

And Sand from Gravel!

After **#22 Gravel**

- Collect 16× **Sand**

Millstone Quicreate



+2 other recipes in JEI

Ingredient	Where it comes from
Gravel	Made in **#22 Gravel**

Item	Also from
Sand	Chest loot in the Cursed Pyramid (Mars), 59% per chest Chest loot in the Desert Site (Mars), 59% per chest



#28 Windmill Bearing

Did you know Sails stick without glue!

After **#18 Hand Crank**

- Collect 1× **Windmill Bearing**

> Windmills produce more power for every 8 sails, and each time they get stronger they spin faster!

Crafting Table create



Ingredient	Where it comes from
any Acacia Slab	Crafting Table: 3× Acacia Planks → 6
any Andesite	Made in **#10 andesite** Mechanical Extruder with Block of Andesite Alloy below (Lava + Water)
Shaft	Made in **#17 Shafts & Cogwheels**



#29 Sails

Windmills Just Work

After **#28 Windmill Bearing**

- Collect 8× **Windmill Sail Frame**

> Sail Frames work just like sails and need no wool.
> Every 8 sails adds power and speed.

Crafting Table (shapeless) create



Ingredient	Where it comes from
Windmill Sail	Crafting Table: any Black Wool + 2× Stick + Andesite Alloy → 2

#30 Andesite Mesh

Different meshes get more materials!

After **#24 Mesh**, **#25 Iron**

- Collect 1× **Andesite Mesh**

Crafting Table pack script



Ingredient	Where it comes from
Andesite Alloy	Made in **#15 Andesite Alloy**
Iron Nugget	Made in **#25 Iron** Sift Sand with Andesite Mesh: **50%** Sift Sand with String Mesh: **25%**
String Mesh	Made in **#24 Mesh**

#31 Zinc

Sift sand with an andesite mesh

After **#30 Andesite Mesh**

- Collect 27× **Zinc Nugget**

Item	Where it comes from
Zinc Nugget	Sift Sand with Zinc Mesh: **35%** Sift Sand with Andesite Mesh: **25%** Open a Light Kelpy Ball (fishing): 16.3% per ball

Or make it:

Encased Fan + Water (washing) create



+6 other recipes in JEI

Ingredient	Where it comes from
Crushed Raw Zinc	Sift Sand with Brass Mesh: **10%**

#32 Rubber

Rubber is a substitute for kelp in create recipes!

After **#30 Andesite Mesh**

- Collect 8× **Rubber Ball**

> Sift sand in a waterlogged sifter with an andesite mesh or better.

Item	Where it comes from
Rubber Ball	Sift Sand with Brass Mesh (waterlogged): **60%** Sift Sand with Zinc Mesh (waterlogged): **40%** Sift Sand with Andesite Mesh (waterlogged): **20%**

#33 Dust

No one knows where dust comes from...

After #27 Sand

- Collect 16× **Dust block**

Millstone Quicreate



Ingredient	Where it comes from
Sand	Made in #27 Sand Chest loot in the Cursed Pyramid (Mars), 59% per chest Chest loot in the Desert Site (Mars), 59% per chest

#34 Redstone

Sift dust with an andesite mesh

After #30 Andesite Mesh, #33 Dust

- Collect 16× **Redstone Dust**

Item	Where it comes from
Redstone Dust	Sift Dust block with Zinc Mesh: **40%** Sift Dust block with Andesite Mesh: **20%** Sift Dust block with Brass Mesh: **20%** The Prowler in the Ancient Factory (Venus), 100% per kill The Watcher in the Ancient Factory (Venus), 100% per kill Chest loot in the Lunar Tower (The Moon), 60.3% per chest

Or make it:

Crafting Table (shapeless) vanilla



+11 other recipes in JEI

Ingredient	Where it comes from
Block of Redstone	Crafting Table: 9× Redstone Dust

#35 Zinc Mesh

After #31 Zinc, #25 Iron

- Collect 1× **Zinc Mesh**

Crafting Table pack script



Ingredient	Where it comes from
Zinc Ingot	Furnace: Crushed Raw Zinc
Iron Ingot	Open a Medium Kelpy Ball (fishing): 20.5% per ball Furnace: Crushed Raw Iron Iron Golem in the Desert Occupied Village (Mars), 100% per kill The Prowler in the Ancient Factory (Venus), 100% per kill
Andesite Mesh	Made in #30 Andesite Mesh

#36 Gold

Sift sand with a zinc mesh

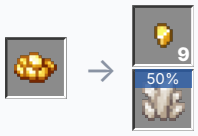
After #35 Zinc Mesh

- Collect 27× Gold Nugget

Item	Where it comes from
Gold Nugget	Sift Sand with Brass Mesh: **50%** Sift Sand with Zinc Mesh: **20%** Open a Light Kelpy Ball (fishing): 16.3% per ball Chest loot in the Sunken City (Glacio), 37.3% per chest Zombified Piglin drops it (100% per kill; where it spawns is not in the data files)

Or make it:

Encased Fan + Water (washing) create



+14 other recipes in JEI

Ingredient	Where it comes from
Crushed Raw Gold	Crushing Wheels: Raw Gold

#37 Copper

Sift gravel with a zinc mesh

After #35 Zinc Mesh

- Collect 27× Copper Nugget

Item	Where it comes from
Copper Nugget	Sift Gravel with Zinc Mesh: **35%** Open a Light Kelpy Ball (fishing): 16.3% per ball

Or make it:

Encased Fan + Water (washing) create



+5 other recipes in JEI

Ingredient	Where it comes from
Crushed Raw Copper	Sift Gravel with Brass Mesh: **12.5%**

#38 Brass Mesh

After #35 Zinc Mesh, #36 Gold, #63 Brass

- Collect 1× **Brass Mesh**

> Needs brass ingots from chapter 1.2.

Crafting Table pack script



Ingredient	Where it comes from
Brass Ingot	Covered later in #63 Brass
Gold Ingot	Open a Medium Kelpy Ball (fishing): 13.1% per ball Furnace: Crushed Raw Gold Chest loot in the Moon Dungeon (The Moon), 22.6% per chest Chest loot in the Ruined Citadel (The Moon), 54% per chest
Zinc Mesh	Made in #35 Zinc Mesh

#39 Powdered Obsidian

Sift dust with a brass mesh

After #38 Brass Mesh

- Collect 8× **Powdered Obsidian**

> Compact 4 in a superheated basin for obsidian, or use it for Sturdy Sheets.

Item	Where it comes from
Powdered Obsidian	Sift Dust block with Advanced Brass Mesh: **50%** Sift Dust block with Brass Mesh: **20%**

#40 Quartz

Sift gravel with a brass mesh

After #38 Brass Mesh

- Collect 16× **Nether Quartz**

Item	Where it comes from
Nether Quartz	Sift Soul Sand with Advanced Brass Mesh: **53.2%** Sift Gravel with Advanced Brass Mesh: **20%** Sift Soul Sand with Brass Mesh: **10%**

Or make it:

Encased Fan + Water (washing) create



+10 other recipes in JEI

Ingredient	Where it comes from
Soul Sand	Millstone: Soul Soil Nether island: 3× Soul Sand

#41 Thorium

Sift gravel with a brass mesh (2.5%)

After **#38 Brass Mesh**

- Collect 4× **Thorium**

Item	Where it comes from
Thorium	Sift Gravel with Advanced Brass Mesh: **10%** Sift Gravel with Brass Mesh: **2.5%**

Or make it:

Mechanical Mixer + Basin create-new-age



Ingredient	Where it comes from
any Andesite	Made in **#10 andesite** Mechanical Extruder with Block of Andesite Alloy below (Lava + Water)
Clay	Crafting Table: 4× Clay Ball
1000 mB Water	

#42 Emeralds

Sift sand with a brass mesh (1%)

After **#38 Brass Mesh**

- Collect 4× **Emerald**

> You will need a few for Emerald Circuits in chapter 1.3.

Item	Where it comes from
Emerald	Sift Sand with Advanced Brass Mesh: **5%** Sift Sand with Brass Mesh: **1%** Chest loot in the Lunar Tower (The Moon), 60.3% per chest Chest loot in the Lunar Tower (The Moon), 20.9% per chest Chest loot in the Lunarian Village (The Moon), 60.3% per chest Chest loot in the Lunarian Village (The Moon), 20.9% per chest

Or make it:

Crafting Table (shapeless) vanilla



+8 other recipes in JEI

Ingredient	Where it comes from
Block of Emerald	Crafting Table: 9× Emerald



1.2: NEW AGE · STEPS #43–#77

New Age

Blaze burners are out of reach, so heat comes from Create: New Age solar heating plates and heaters. Heated mixing makes brass; a Generator Coil spun next to magnetite makes electricity for the Energiser. The Mechanical Extruder turns lava and water into stone types, and the block you put under it changes what it makes (Appendix B).



#43 Mechanical Press

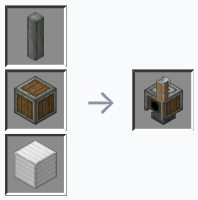
Ingots into sheets

After **#16 Andesite Casing**, **#25 Iron**

- Collect 1× **Mechanical Press**

> Needs a block of iron.

Crafting Table create



Ingredient	Where it comes from
Shaft	Made in **#17 Shafts & Cogwheels**
Andesite Casing	Made in **#16 Andesite Casing**
Block of Iron	Crafting Table: 9× Iron Ingot



#44 Heat Generation

Step 1: Creating Heat

After **#37 Copper**

- **Read this** (tick it in the quest book)

> Blaze burners are generally inaccessible, so solar heat is the replacement for heating operations.



#45 Electricity

Step 2: Tools for Electricity

- **Read this** (tick it in the quest book)

#46 Crushing is The New Sifting

- Collect 2× **Crushing Wheel**

> Crush crimsite, veridium, asurine and ochrum for crushed ores.

Mechanical Crafters create



Ingredient	Where it comes from
Andesite Alloy	Made in **#15 Andesite Alloy**
any Planks	Crafting Table (shapeless): any Oak Log → 4
any Andesite	Made in **#10 andesite** Mechanical Extruder with Block of Andesite Alloy below (Lava + Water)

#47 Sheet Metal

Press ingots on a depot or belt

After #36 Gold, #37 Copper, #43 Mechanical Press

- Collect 8× **Iron Sheet**
- Collect 8× **Copper Sheet**
- Collect 4× **Golden Sheet**

Mechanical Press create



Compressor Quicreate



+2 other recipes in JEI

Ingredient	Where it comes from
Iron Ingot	Open a Medium Kelpy Ball (fishing): 20.5% per ball Crushing Wheels: Iron Horse Armor → 2 Iron Golem in the Desert Occupied Village (Mars), 100% per kill The Prowler in the Ancient Factory (Venus), 100% per kill

Mechanical Press create



Ingredient	Where it comes from
Copper Ingot	Furnace: Crushed Raw Copper Chest loot in the Acropolis (Glacio), 28.6% per chest Drowned drops it (11% per kill; where it spawns is not in the data files)

Mechanical Press create



+2 other recipes in JEI

Ingredient	Where it comes from
Gold Ingot	Open a Medium Kelpy Ball (fishing): 13.1% per ball Crushing Wheels: Golden Horse Armor → 2 Chest loot in the Moon Dungeon (The Moon), 22.6% per chest Chest loot in the Ruined Citadel (The Moon), 54% per chest

#48 Blaze Burner?

Hey you aren't breaking the intended progression are you?

- Collect 1× **Empty Blaze Burner**

This quest exists to catch a blaze burner made outside the intended route. Heat for mixing comes from New Age heaters and solar plates (the next steps).

Crafting Table create



Ingredient	Where it comes from
any Iron Plate	Made in #47 Sheet Metal
Netherrack	Mechanical Extruder with Nether Bricks below (Lava + Water) Nether island: 63× Netherrack Nether island: 29× Crimson Nylium (drops Netherrack) Nether island: 29× Warped Nylium (drops Netherrack)

#49 Solar Plate

Free heat energy! Sometimes...

After #47 Sheet Metal

- Collect 2× **Basic Solar Heating Plate**

Crafting Table pack script



Ingredient	Where it comes from
Zinc Ingot	Furnace: Crushed Raw Zinc
Glass	Encased Fan + Water (washing): any Black Stained Glass
Copper Sheet	Made in #47 Sheet Metal
Golden Sheet	Made in #47 Sheet Metal
Iron Sheet	Made in #47 Sheet Metal

#50 Heater

Whats a Blaze Burner!

After #49 Solar Plate

- Collect 1× **Heater**

Crafting Table pack script



Ingredient	Where it comes from
Copper Sheet	Made in #47 Sheet Metal
Zinc Ingot	Furnace: Crushed Raw Zinc
Cauldron	Crafting Table: 7× Iron Ingot
Furnace	Made in #9 furnace

#51 Mechanical Saw

Cuts sheets into wires

After #43 Mechanical Press

- Collect 1× **Mechanical Saw**

Crafting Table create



Ingredient	Where it comes from
any Iron Plate	Made in #47 Sheet Metal
Iron Ingot	Open a Medium Kelpy Ball (fishing): 20.5% per ball Crushing Wheels: Iron Horse Armor → 2 Iron Golem in the Desert Occupied Village (Mars), 100% per kill The Prowler in the Ancient Factory (Venus), 100% per kill
Andesite Casing	Made in #16 Andesite Casing

#52 Magnetite

It's Magnetic!

After #34 Redstone

- Collect 8× **Magnetite Block**

> Magnetite should be placed against a spinning generator coil to create electricity. up to 12 blocks can be used per coil, and magnetite can be upgraded to create more efficient systems.

Crafting Table Quicreate



Ingredient	Where it comes from
Andesite Alloy	Made in #15 Andesite Alloy
Iron Sheet	Made in #47 Sheet Metal
Redstone Dust	Made in #34 Redstone Sift Dust block with Zinc Mesh: **40%** Sift Dust block with Andesite Mesh: **20%**



#53 Carbon Brush

Absorbs electricity generated by a Coil!

- Collect 1× **Carbon Brushes**

> The Carbon Brush is how you extract electricity from your Generator Coil. Simply attach it directly to the end of the coil and use a connector to transfer electricity.

Crafting Table pack script



Ingredient	Where it comes from
Zinc Ingot	Furnace: Crushed Raw Zinc
Copper Sheet	Made in **#47 Sheet Metal**
Coal	Sift Gravel with Andesite Mesh: **30%** Sift Venus Sand with Advanced Brass Mesh: **25%** Sift Gravel with Zinc Mesh: **20%** Crafting Table (shapeless): Block of Coal → 9
Shaft	Made in **#17 Shafts & Cogwheels**



#54 Basin

After **#50 Heater**

- Collect 1× **Basin**

Crafting Table create



Ingredient	Where it comes from
Andesite Alloy	Made in **#15 Andesite Alloy**

#55 Heat Pipes

Evenly move your heat

After #50 Heater

- Collect 8× **Heat Pipe**

Crafting Table pack script



Ingredient	Where it comes from
Zinc Ingot	Furnace: Crushed Raw Zinc
Copper Sheet	Made in #47 Sheet Metal

#56 Heat Pumps

Push your heat around

After #55 Heat Pipes

- Collect 2× **Heat Pump**

Crafting Table pack script



Ingredient	Where it comes from
Zinc Ingot	Furnace: Crushed Raw Zinc
Copper Sheet	Made in #47 Sheet Metal

#57 Stirling Engine

Convert Heat, into rotation!

After #50 Heater

- Collect 1× **Stirling Engine**

Crafting Table pack script



Ingredient	Where it comes from
Copper Sheet	Made in #47 Sheet Metal
Zinc Ingot	Furnace: Crushed Raw Zinc
Shaft	Made in #17 Shafts & Cogwheels

#58 Wires

Cut copper sheets with a saw

- Collect 16× **Copper Wire**

Mechanical Saw create-new-age



Crafting Table (shapeless) create-new-age



Ingredient	Where it comes from
Copper Sheet	Made in #47 Sheet Metal

#59 Connector

After #53 Carbon Brush

- Collect 8× **Electrical Connector**

Crafting Table pack script



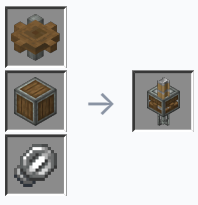
Ingredient	Where it comes from
Zinc Ingot	Furnace: Crushed Raw Zinc
Copper Sheet	Made in #47 Sheet Metal

#60 Mixer

After #54 Basin

- Collect 1× **Mechanical Mixer**

Crafting Table create



Ingredient	Where it comes from
Cogwheel	Made in #17 Shafts & Cogwheels
Andesite Casing	Made in #16 Andesite Casing
Whisk	Crafting Table: 2× Andesite Alloy + 5× any Iron Plate

#61 Lava

Superheated mixing turns cobble into lava

After #60 Mixer, #56 Heat Pumps

- Collect 1× **Lava Bucket**

Item	Where it comes from
Lava Bucket	Starting island: 1× in the starting chest

#62 Energizer

After #59 Connector

- Collect 1× **Basic Energiser**

Crafting Table (shapeless) create-new-age



Ingredient	Where it comes from
Andesite Casing	Made in #16 Andesite Casing
Lightning Rod	Crafting Table: 3× Copper Ingot

#63 Brass

Copper + zinc in a heated mixer

After #60 Mixer

- Collect 16× **Brass Ingot**

Mechanical Mixer + Basin Heated create



+3 other recipes in JEI

Ingredient	Where it comes from
Copper Ingot	Furnace: Crushed Raw Copper Chest loot in the Acropolis (Glacio), 28.6% per chest Drowned drops it (11% per kill; where it spawns is not in the data files)
Zinc Ingot	Furnace: Crushed Raw Zinc

#64 Brass Casing

Brass on a stripped log

After #63 Brass

- Collect 8× **Brass Casing**

Use item on block (hand or Deployer) create



Use item on block (hand or Deployer) create



Ingredient	Where it comes from
any Stripped Oak Log	Strip a log or wood block by right-clicking it with an axe. (game mechanic)
Brass Ingot	Made in #63 Brass

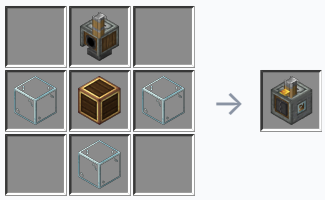
#65 Mechanical Extruder

Generator for Cobble and more!

- Collect 1× **Mechanical Extruder**

> The mechanical extruder will produce blocks nonstop. Stone is made by default, but the ouput can be changed by placing particular blocks below it!

Crafting Table pack script



Ingredient	Where it comes from
Mechanical Press	Made in #43 Mechanical Press
Glass	Encased Fan + Water (washing): any Black Stained Glass
Brass Casing	Made in #64 Brass Casing

#66 Crimsite

Easy Iron (and redstone)

After #65 Mechanical Extruder

- Collect 32× **Crimsite**

> Put a block of iron under the extruder.

Item	Where it comes from
Crimsite	Mechanical Extruder with Block of Iron below (Lava + Water)

#67 Veridium

Consistent Copper (and clay)

After #65 Mechanical Extruder

- Collect 32× **Veridium**

> Put a block of copper under the extruder.

Item	Where it comes from
Veridium	Mechanical Extruder with Block of Copper below (Lava + Water) Mechanical Extruder with Exposed Copper below (Lava + Water) Mechanical Extruder with Oxidized Copper below (Lava + Water)

#68 Asurine

Sufficient Zinc (and gunpowder)

After #65 Mechanical Extruder

- Collect 32× **Asurine**

> Put a block of zinc under the extruder.

Item	Where it comes from
Asurine	Mechanical Extruder with Block of Zinc below (Lava + Water)

#69 Ochrum

G O L D (and quartz)

After #65 Mechanical Extruder

- Collect 32× **Ochrum**

> Put a block of gold under the extruder.

Item	Where it comes from
Ochrum	Mechanical Extruder with Block of Gold below (Lava + Water)

#70 Overcharged Iron

After #62 Energizer

- Collect 8× **Overcharged Iron**

Energiser 1000 energy — create-new-age



Ingredient	Where it comes from
Iron Ingot	Open a Medium Kelpy Ball (fishing): 20.5% per ball Crushing Wheels: Iron Horse Armor → 2 Iron Golem in the Desert Occupied Village (Mars), 100% per kill The Prowler in the Ancient Factory (Venus), 100% per kill

#71 Blaze Powder

Energise gunpowder

After #62 Energizer

- Collect 8× Blaze Powder

Energiser 2000 energy — Quicreate



Crafting Table (shapeless) — vanilla



+2 other recipes in JEI

Ingredient	Where it comes from
Gunpowder	Sift Dust block with String Mesh: **25%** Sift Dust block with Andesite Mesh: **15%** Sift Dust block with Zinc Mesh: **5%** Chest loot in the Ruined Citadel (The Moon), 57.8% per chest

#72 Electron Tubes

Rose quartz, polished, on an iron sheet

After #40 Quartz, #34 Redstone, #47 Sheet Metal

- Collect 8× Electron Tube

> Rose quartz = quartz + 8 redstone. Polish it with sandpaper.

Crafting Table — create



+1 other recipe in JEI

Ingredient	Where it comes from
Polished Rose Quartz	Sand Paper (hold and use): Rose Quartz
any Iron Plate	Made in #47 Sheet Metal

#73 Mechanical Crafters

The Generator Coil needs a 5×5 grid (21 crafters)

- Collect 21× Mechanical Crafter

Crafting Table — create



Ingredient	Where it comes from
Electron Tube	Made in #72 Electron Tubes
Brass Casing	Made in #64 Brass Casing
Crafting Table	Made in #3 Get Crafting Table

#74 Generator Coil

Generates power when it spins near a magnet

After **#73 Mechanical Crafters**, **#58 Wires**

- Collect 1× **Generator Coil**

> The Generator Coil is spun while touching Magnetite to generate electricity. The more Magnetite you use, the more Electricity. A Coil takes a lot of power to spin, so stronger Magnetites will create electricity more efficiently.

Mechanical Crafters Quicreate



Ingredient	Where it comes from
Zinc Ingot	Furnace: Crushed Raw Zinc
Copper Wire	Made in **#58 Wires**
Iron Ingot	Open a Medium Kelpy Ball (fishing): 20.5% per ball Crushing Wheels: Iron Horse Armor → 2 Iron Golem in the Desert Occupied Village (Mars), 100% per kill The Prowler in the Ancient Factory (Venus), 100% per kill
Block of Zinc	Crafting Table: 9× Zinc Ingot

#75 Overcharged Gold

After **#62 Energizer**

- Collect 8× **Overcharged Gold**

Energiser 2000 energy create-new-age



Ingredient	Where it comes from
Gold Ingot	Open a Medium Kelpy Ball (fishing): 13.1% per ball Crushing Wheels: Golden Horse Armor → 2 Chest loot in the Moon Dungeon (The Moon), 22.6% per chest Chest loot in the Ruined Citadel (The Moon), 54% per chest

#76 Deployer

Needed for circuits and sequenced assembly

After #64 Brass Casing, #72 Electron Tubes

- Collect 1× **Deployer**

Crafting Table create



Ingredient	Where it comes from
Electron Tube	Made in #72 Electron Tubes
Andesite Casing	Made in #16 Andesite Casing
Brass Hand	Crafting Table: Andesite Alloy + 4× Brass Sheet

#77 Precision Mechanism

Sequenced assembly on a golden sheet

After #76 Deployer

- Collect 4× **Precision Mechanism**

Sequenced Assembly create



Ingredient	Where it comes from
Golden Sheet	Made in #47 Sheet Metal
Cogwheel	Made in #17 Shafts & Cogwheels
Large Cogwheel	Crafting Table (shapeless): Shaft + 2× any Planks
Iron Nugget	Made in #25 Iron Sift Sand with Andesite Mesh: **50%** Sift Sand with String Mesh: **25%**



1.3: IMPROVEMENTS · STEPS #78–#98

Improvements

Circuits are the backbone of the rest of the pack. Superheated heat (from strong solar or a reactor) unlocks steel, which the rocket needs.



#78 Steel

Iron + coal in a superheated mixer

After #61 Lava, #60 Mixer

- Collect 32× **Steel Ingot**

> Superheated heat comes from strong solar or nuclear heaters.

Mechanical Mixer + Basin Superheated Quicreate



Mechanical Mixer + Basin Superheated Quicreate



+3 other recipes in JEI

Ingredient	Where it comes from
Iron Ingot	Open a Medium Kelpy Ball (fishing): 20.5% per ball Crushing Wheels: Iron Horse Armor → 2 Iron Golem in the Desert Occupied Village (Mars), 100% per kill The Prowler in the Ancient Factory (Venus), 100% per kill
Charcoal	Made in #14 Charcoal



#79 Motors

Motors are an easy way to put you SU anywhere!

After #52 Magnetite, #47 Sheet Metal

- Collect 1× **Basic Motor**

Crafting Table pack script



Ingredient	Where it comes from
Iron Sheet	Made in #47 Sheet Metal
Andesite Casing	Made in #16 Andesite Casing
Magnetite Block	Made in #52 Magnetite
Shaft	Made in #17 Shafts & Cogwheels

#80 Nuclear Heat

Crushed Thorium can provide Radioactive Thorium

After #41 Thorium, #46 Crushing is The New Sifting

- Collect 4× **Radioactive Thorium**

Crushing Wheels create-new-age



Ingredient	Where it comes from
Thorium	Made in #41 Thorium Sift Gravel with Advanced Brass Mesh: **10%** Sift Gravel with Brass Mesh: **2.5%**

#81 Redstone Magnet

Better magnets have Better Efficiency.

After #70 Overcharged Iron, #52 Magnetite

- Collect 4× **Redstone Magnet**

Crafting Table pack script



Ingredient	Where it comes from
Magnetite Block	Made in #52 Magnetite
Overcharged Iron Sheet	Energiser (1000 energy): any Iron Plate
Redstone Dust	Made in #34 Redstone Sift Dust block with Zinc Mesh: **40%** Sift Dust block with Andesite Mesh: **20%**

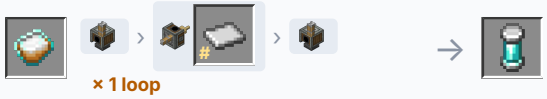
#82 Nuclear Fuel

Heat! at night 😎.

After #80 Nuclear Heat, #43 Mechanical Press, #76 Deployer

- Collect 4× **Nuclear Fuel**

Sequenced Assembly create-new-age



Ingredient	Where it comes from
Radioactive Thorium	Made in #80 Nuclear Heat
any Iron Plate	Made in #47 Sheet Metal

#83 Plastic

Card bits or oil, plus blaze powder

After #71 Blaze Powder

- Collect 8× **Plastic**

> Compact card bits + blaze powder (heated), or mix oil + blaze powder (superheated).

Mechanical Press + Basin (compacting) Heated — Quicreate



Mechanical Mixer + Basin Superheated — Quicreate



Ingredient	Where it comes from
Card Bits	Mechanical Saw: any Allay Card → 2
Blaze Powder	Made in #71 Blaze Powder

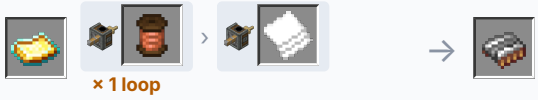
#84 Blank Circuit

Overcharged golden sheet + copper wire + plastic

After #83 Plastic, #75 Overcharged Gold, #58 Wires, #76 Deployer

- Collect 8× **Blank Circuit**

Sequenced Assembly — Quicreate



Ingredient	Where it comes from
Overcharged Golden Sheet	Energiser (2000 energy): Golden Sheet
Copper Wire	Made in #58 Wires
Plastic	Made in #83 Plastic

#85 Copper Circuit

Overcharged iron wire and quartz, three times

After #84 Blank Circuit, #70 Overcharged Iron, #40 Quartz

- Collect 8× **Copper Circuit**

Sequenced Assembly — Quicreate



Ingredient	Where it comes from
Incomplete Copper Circuit	Deployer: Blank Circuit + Copper Sheet
Overcharged Iron Wire	Mechanical Saw: Overcharged Iron Sheet → 4
Nether Quartz	Made in #40 Quartz Sift Soul Sand with Advanced Brass Mesh: **53.2%** Sift Gravel with Advanced Brass Mesh: **20%**

#86 Motor Extension

The motor extension will multiply the maximum SU output of a motor

After #79 Motors, #85 Copper Circuit, #81 Redstone Magnet

- Collect 1× **Basic Motor Extension**

Crafting Table pack script



Ingredient	Where it comes from
Iron Sheet	Made in #47 Sheet Metal
Copper Circuit	Made in #85 Copper Circuit
Redstone Magnet	Made in #81 Redstone Magnet

#87 Moon Stuff

Copper Circuits are also used for space related things!

After #85 Copper Circuit

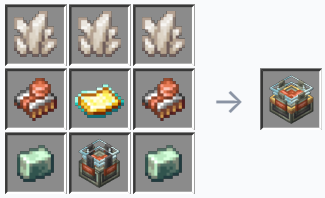
- **Read this** (tick it in the quest book)

#88 Advanced Solar Plate

After #85 Copper Circuit, #75 Overcharged Gold

- Collect 2× **Advanced Solar Heating Plate**

Crafting Table pack script



Ingredient	Where it comes from
Nether Quartz	Made in #40 Quartz Sift Soul Sand with Advanced Brass Mesh: **53.2%** Sift Gravel with Advanced Brass Mesh: **20%**
Copper Circuit	Made in #85 Copper Circuit
Overcharged Golden Sheet	Energiser (2000 energy): Golden Sheet
Zinc Ingot	Furnace: Crushed Raw Zinc
Basic Solar Heating Plate	Made in #49 Solar Plate

#89 Layered Magnet

After #81 Redstone Magnet, #75 Overcharged Gold

- Collect 2× **Layered Magnet**

Crafting Table pack script



Ingredient	Where it comes from
Redstone Magnet	Made in #81 Redstone Magnet
Overcharged Golden Sheet	Energiser (2000 energy): Golden Sheet
Overcharged Iron Sheet	Energiser (1000 energy): any Iron Plate

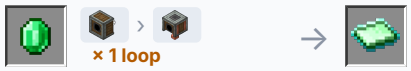
#90 Emerald Wafer

Cut an emerald, then energise it

After #42 Emeralds, #51 Mechanical Saw, #62 Energizer

- Collect 2× **Emerald Wafer**

Sequenced Assembly Quicreate



Ingredient	Where it comes from
Emerald	Made in #42 Emeralds Sift Sand with Advanced Brass Mesh: **5%** Sift Sand with Brass Mesh: **1%**

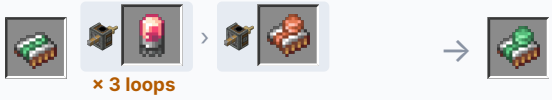
#91 Emerald Circuit

Electron tubes and copper circuits on a blank circuit

After #85 Copper Circuit, #90 Emerald Wafer, #72 Electron Tubes

- Collect 4× **Emerald Circuit**

Sequenced Assembly Quicreate



Ingredient	Where it comes from
Incomplete Emerald Circuit	Deployer: Blank Circuit + Emerald Wafer
Electron Tube	Made in #72 Electron Tubes
Copper Circuit	Made in #85 Copper Circuit

#92 Strong Motor

After #91 Emerald Circuit, #89 Layered Magnet, #64 Brass Casing, #73 Mechanical Crafters

- Collect 1× **Advanced Motor**

Mechanical Crafters Quicreate



Ingredient	Where it comes from
Overcharged Iron Sheet	Energiser (1000 energy): any Iron Plate
Overcharged Golden Sheet	Energiser (2000 energy): Golden Sheet
Brass Casing	Made in #64 Brass Casing
Emerald Circuit	Made in #91 Emerald Circuit
Layered Magnet	Made in #89 Layered Magnet
Shaft	Made in #17 Shafts & Cogwheels

#93 Storage Terminals

A Storage Terminal lets you access mass storage and vaults like a chest!

After #91 Emerald Circuit

- Collect 1× **Storage Terminal**

Crafting Table Quicreate



Ingredient	Where it comes from
any Acacia Slab	Crafting Table: 3× Acacia Planks → 6
Emerald Circuit	Made in #91 Emerald Circuit
List Filter	Crafting Table (shapeless): List Filter

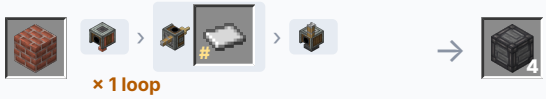
#94 Reactor Case

Energise bricks, add an iron sheet, press

After #62 Energizer

- Collect 16× **Reactor Casing**

Sequenced Assembly create-new-age



Ingredient	Where it comes from
Bricks	Crafting Table: 4× Brick
any Iron Plate	Made in #47 Sheet Metal

#95 Fuel Inserter

Connects to the reactor rods to load them

After #91 Emerald Circuit, #94 Reactor Case

- Collect 1× **Reactor Fuel Acceptor**

Crafting Table pack script



Ingredient	Where it comes from
Reactor Casing	Made in #94 Reactor Case
Andesite Funnel	Crafting Table: Andesite Alloy + Rubber Ball → 2
Steel Plate	Covered later in #99 Steel Plates
Emerald Circuit	Made in #91 Emerald Circuit

#96 Reactor Rod

Make sure this is covered on all sides with reactor parts!

After #94 Reactor Case, #82 Nuclear Fuel, #73 Mechanical Crafters

- Collect 2× **Reactor Rod**

Mechanical Crafters create-new-age



Ingredient	Where it comes from
Reactor Casing	Made in #94 Reactor Case
Golden Sheet	Made in #47 Sheet Metal
Reactor Glass	Crafting Table: 4× Reactor Casing + 5× any Framed Glass
Nuclear Fuel	Made in #82 Nuclear Fuel

#97 Heat Vent

Export the Heat from the reaction, or else

After #91 Emerald Circuit, #94 Reactor Case, #56 Heat Pumps

- Collect 2× **Reactor Heat Vent**

Crafting Table pack script



Ingredient	Where it comes from
Reactor Casing	Made in #94 Reactor Case
Heat Pump	Made in #56 Heat Pumps
Emerald Circuit	Made in #91 Emerald Circuit
Steel Plate	Covered later in #99 Steel Plates

#98 Advanced Energizer

After #91 Emerald Circuit

- Collect 1× **Advanced Energiser**

Crafting Table pack script



Ingredient	Where it comes from
Overcharged Golden Sheet	Energiser (2000 energy): Golden Sheet
Emerald Circuit	Made in #91 Emerald Circuit
Basic Energiser	Made in #62 Energizer
Copper Sheet	Made in #47 Sheet Metal
Lightning Rod	Crafting Table: 3× Copper Ingot



1.4: SPACE · STEPS #99–#116

Space

Everything for the tier 1 rocket. The rocket itself is assembled in the NASA Workbench.



#99 Steel Plates

Press steel ingots

After #78 Steel, #43 Mechanical Press

- Collect 32× Steel Plate

Mechanical Press ad astra



Compressor Quicreate



+1 other recipe in JEI

Ingredient	Where it comes from
Steel Ingot	Made in #78 Steel



#100 Nose Cone & Fins

After #99 Steel Plates

- Collect 1× Rocket Nose Cone
- Collect 4× Rocket Fin

Crafting Table pack script



Ingredient	Where it comes from
Lightning Rod	Crafting Table: 3× Copper Ingot
Steel Plate	Made in #99 Steel Plates
Glass	Encased Fan + Water (washing): any Black Stained Glass

Crafting Table pack script



Ingredient	Where it comes from
Steel Rod	Crafting Table: 2× Steel Plate → 4
Steel Plate	Made in #99 Steel Plates

#101 Space Helmet

You cannot breath on the moon typically

- Collect 1× **Space Helmet**

Crafting Table pack script



Ingredient	Where it comes from
Steel Plate	Made in **#99 Steel Plates**
Glass	Encased Fan + Water (washing): any Black Stained Glass

Item	Also from
Space Helmet	Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest

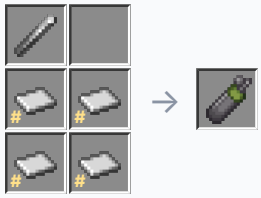
#102 Gas Tanks

Iron plates and an iron rod

After **#47 Sheet Metal**

- Collect 6× **Gas Tank**

Crafting Table ad astra



Ingredient	Where it comes from
Iron Rod	Crafting Table: 2× any Iron Sheet → 4
any Iron Sheet	Made in **#47 Sheet Metal**

Item	Also from
Gas Tank	Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest

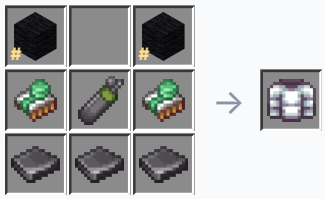
#103 Space Suit

Has internal tanks to hold the oxygen you need!

After #102 Gas Tanks, #91 Emerald Circuit

- Collect 1× **Space Suit**

Crafting Table Quicreate



Ingredient	Where it comes from
any Black Wool	Crafting Table (shapeless): Black Dye + Blue Wool
Emerald Circuit	Made in #91 Emerald Circuit
Gas Tank	Made in #102 Gas Tanks Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest
Steel Plate	Made in #99 Steel Plates

Item	Also from
Space Suit	Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest

#104 Large Gas Tank

Bring extra oxygen to refill your suit

After #99 Steel Plates, #102 Gas Tanks

- Collect 1× **Large Gas Tank**

Crafting Table ad astra



Ingredient	Where it comes from
Steel Rod	Crafting Table: 2× Steel Plate → 4
Steel Plate	Made in #99 Steel Plates
Gas Tank	Made in #102 Gas Tanks Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest

#105 Steel Tanks

After #99 Steel Plates, #102 Gas Tanks

- Collect 2× **Steel Tank**

Crafting Table pack script



Ingredient	Where it comes from
Steel Plate	Made in #99 Steel Plates
Steel Rod	Crafting Table: 2× Steel Plate → 4
Gas Tank	Made in #102 Gas Tanks Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest

#106 Rocket Fan

Steel around a propeller

After #99 Steel Plates

- Collect 2× **Steel Rocket Propeller**

Crafting Table pack script



Ingredient	Where it comes from
Steel Rod	Crafting Table: 2× Steel Plate → 4
Steel Plate	Made in #99 Steel Plates
Propeller	Crafting Table: 4× any Iron Plate + Andesite Alloy

#107 Space Pants

You do need the whole set

- Collect 1× **Space Pants**

Crafting Table ad astra



Ingredient	Where it comes from
Steel Plate	Made in **#99 Steel Plates**
any Black Wool	Crafting Table (shapeless): Black Dye + Blue Wool

Item	Also from
Space Pants	Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest

#108 Oxygen Loader

Create Oxygen from Water

After **#102 Gas Tanks**, **#82 Nuclear Fuel**, **#85 Copper Circuit**

- Collect 1× **Oxygen Loader**

Crafting Table pack script



Ingredient	Where it comes from
Reactor Casing	Made in **#94 Reactor Case**
Copper Circuit	Made in **#85 Copper Circuit**
Gas Tank	Made in **#102 Gas Tanks** Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest
Nuclear Fuel	Made in **#82 Nuclear Fuel**

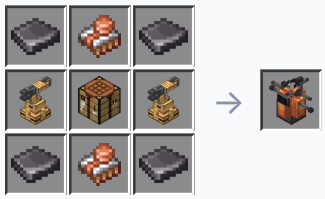
#109 Nasa Workbench

Used to craft the rocket

After #99 Steel Plates, #85 Copper Circuit

- Collect 1× Rocket Assembly Bench

Crafting Table pack script

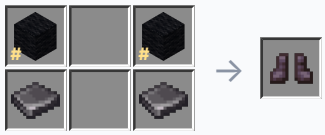


Ingredient	Where it comes from
Steel Plate	Made in #99 Steel Plates
Copper Circuit	Made in #85 Copper Circuit
Mechanical Arm	Crafting Table: 3× Brass Sheet + Andesite Alloy + Precision Mechanism + Brass Casing
Crafting Table	Made in #3 Get Crafting Table

#110 Space Boots

- Collect 1× Space Boots

Crafting Table ad astra



Ingredient	Where it comes from
any Black Wool	Crafting Table (shapeless): Black Dye + Blue Wool
Steel Plate	Made in #99 Steel Plates

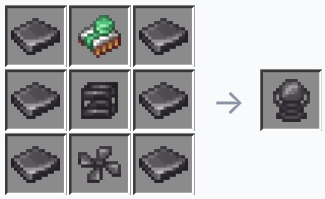
Item	Also from
Space Boots	Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest

#111 Steel Engine

After #106 Rocket Fan, #99 Steel Plates, #91 Emerald Circuit

- Collect 1× **Steel Engine**

Crafting Table pack script



Ingredient	Where it comes from
Steel Plate	Made in #99 Steel Plates
Emerald Circuit	Made in #91 Emerald Circuit
Engine Frame	Crafting Table: 8× Steel Rod + Steel Plate
Steel Rocket Propeller	Made in #106 Rocket Fan

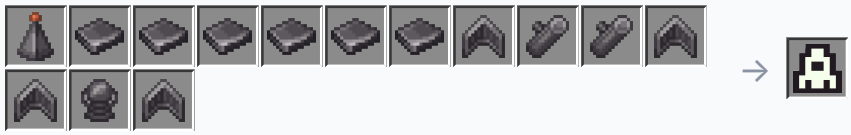
#112 Rocket

Built in the NASA Workbench

After #100 Nose Cone & Fins, #105 Steel Tanks, #111 Steel Engine, #109 Nasa Workbench

- Collect 1× **Tier 1 Rocket**

NASA Workbench Quicreate



Ingredient	Where it comes from
Rocket Nose Cone	Made in #100 Nose Cone & Fins
Steel Plate	Made in #99 Steel Plates
Rocket Fin	Made in #100 Nose Cone & Fins
Steel Tank	Made in #105 Steel Tanks
Steel Engine	Made in #111 Steel Engine

#113 Oil

Rubber balls in a superheated mixer

After #61 Lava, #32 Rubber

- Collect 1× **Oil Bucket**

Item	Where it comes from
Oil Bucket	Fill a bucket with Oil (Spout, or pick it up)

#114 Launch Pad

After #99 Steel Plates

- Collect 1× Launch Pad

Crafting Table ad astra



Ingredient	Where it comes from
Steel Plate	Made in #99 Steel Plates
Steel Rod	Crafting Table: 2× Steel Plate → 4

Item	Also from
Launch Pad	Chest loot in the Lunar Tower (The Moon), 29.7% per chest Chest loot in the Lunarian Village (The Moon), 29.7% per chest

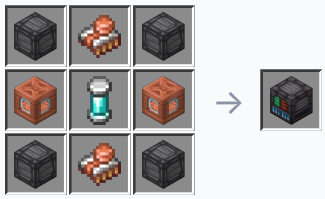
#115 Fuel Refinery

Create Rocket Fuel From Oil

After #113 Oil, #94 Reactor Case, #82 Nuclear Fuel, #85 Copper Circuit

- Collect 1× Fuel Refinery

Crafting Table pack script



Ingredient	Where it comes from
Reactor Casing	Made in #94 Reactor Case
Copper Circuit	Made in #85 Copper Circuit
Fluid Tank	Crafting Table: 2× Copper Sheet + Barrel
Nuclear Fuel	Made in #82 Nuclear Fuel

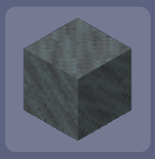
#116 Rocket Fuel

Bring enough fuel to get Home too! Or don't

After #115 Fuel Refinery

- Collect 3× Fuel Bucket

Item	Where it comes from
Fuel Bucket	Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest



CHAPTER 2: THE MOON · STEPS #117–#145

The Moon

The only stronghold in the pack is on the Moon (the stronghold biome tag is set to the lunar wastelands). Kill the Ender Dragon to finish the soft ending, then build a tier 2 rocket for Mars.

The Moon: what is here

Biomes: Lunar Wastelands, Cratered Maria

Terrain: Moon Stone Moon Sand Moon Deepslate

Ore / deposit	Drops	Height	Veins / chunk	Vein size
Moon Cheese Ore	Cheese	6 to 192	9	8
Moon Desh Ore	Raw Desh	-80 to 80	9	9
Moon Ice Shard Ore	Ice Shard	-32 to 32	8	10
Moon Iron Ore	Raw Iron	-24 to 56	10	11
Soul Soil	itself	0 to 100	20	60

Surface features (from the worldgen files):

- **moon crater rare**: Moon Deepslate Moon Sand Moon Ice Shard Ore
- **moon aeronos cave mushroom**: Aeronos Stem Aeronos Cap Shroomlight
- **moon aeronos cave patch**: Aeronos Mushroom
- **moon crater**: Moon Deepslate Moon Sand Moon Cobblestone Moon Ice Shard Ore
- **moon crater small**: Moon Deepslate Moon Sand Moon Ice Shard Ore
- **moon boulder**: Sky Stone Moon Cobblestone

Mob (natural spawns)	Drops per kill
Amethyst Crab	Amethyst Shard 100%, Amethyst Crab Meat 100%, Amethyst Crab Shell 100%
Corrupted Lunarian	no loot table in the files
Star Crawler	Diamond 100%
Enderman	Ender Pearl 100%

Structure (biomes)	Mobs inside	Chest loot highlights
Lunar Tower Lunar Wastelands, Cratered Maria	Lunarian	Moon Sand 89.2% Desh Ingot 86.7% Emerald 60.3% Redstone Dust 60.3% Diamond 41.6% Iron Nugget 38% Launch Pad 29.7% Fuel Bucket 18.2%
Lunarian Village Lunar Wastelands, Cratered Maria	Lunarian	Moon Sand 89.2% Desh Ingot 86.7% Emerald 60.3% Redstone Dust 60.3% Diamond 41.6% Iron Nugget 38% Launch Pad 29.7% Fuel Bucket 18.2%
Moon Dungeon Lunar Wastelands, Cratered Maria	Corrupted Lunarian, Star Crawler	Moon Sand 79.4% Emerald 68.1% Ice Shard 29.9% Diamond 22.6% Redstone Dust 11.9% Ice 52% Cheese 29.9% Raw Iron 29.9%
Ruined Citadel Lunar Wastelands, Cratered Maria	Endermaptera	String 57.8% Redstone Dust 27.4% Diamond 21.5% Emerald 9.1% Bone 57.8% Gunpowder 57.8% Rotten Flesh 57.8% Gold Ingot 54%
Stronghold Lunar Wastelands	–	–

Boss	Found in	Step
Ender Guardian (optional)	Ruined Citadel	#127 Ender Guardian
Ender Dragon	–	#140 Free the End

#117 One Small Step

Land on the Moon

After **#114 Launch Pad**, **#108 Oxygen Loader**, **#110 Space Boots**, **#112 Rocket**, **#116 Rocket Fuel**

- Travel to **Moon**

> Bring fuel for the trip home!

#118 Ruined Citadel

A fortress on the lunar wastelands

After **#117 One Small Step**

- Find the **Ruined Citadel** on The Moon

#119 Ender Pearls

Endermen roam the Moon

After **#117 One Small Step**

- Collect 12× **Ender Pearl**

Item	Where it comes from
Ender Pearl	Sift Crushed end stone with Advanced Brass Mesh: **10%** Enderman (The Moon) drops it, 100% per kill

#120 Moon Cobblestone

Moon cobblestone can be made in a mechanical extruder

After **#117 One Small Step**

- Collect 32× **Moon Cobblestone**

Item	Where it comes from
Moon Cobblestone	Mechanical Extruder with Moon Cobblestone below (Lava + Water) The Moon: mine the Moon Stone terrain The Moon: found in world features (Moon Cobblestone)

#121 Desh

Desh is a minor a resource used for some technologies

After **#117 One Small Step**

- Collect 16× **Desh Ingot**

Furnace ad astra



Crafting Table ad astra



+6 other recipes in JEI

Ingredient	Where it comes from
Raw Desh	Crafting Table (shapeless): Raw Desh Block → 9 The Moon: Moon Desh Ore → Raw Desh (Y -80 to 80) Chest loot in the Moon Dungeon (The Moon), 21.8% per chest Chest loot in the Moon Dungeon (The Moon), 22.6% per chest

Item	Also from
Desh Ingot	Chest loot in the Lunar Tower (The Moon), 86.7% per chest Chest loot in the Lunar Tower (The Moon), 20.9% per chest Chest loot in the Lunarian Village (The Moon), 86.7% per chest Chest loot in the Lunarian Village (The Moon), 20.9% per chest

#122 Diamonds

They're here, but not in ore form

After **#117 One Small Step**

- Collect 8× **Diamond**

> Star crawlers drop them, and the advanced mesh sifts them.

Item	Where it comes from
Diamond	Sift Venus Sand with Advanced Brass Mesh: **5%** Sift Mars Sand with Advanced Brass Mesh: **2.5%** Sift Sand with Advanced Brass Mesh: **1%** Open a Heavy Kelpy Ball (fishing): 15% per ball Mars: Mars Diamond Ore → Diamond (Y -80 to 80) Star Crawler (The Moon) drops it, 100% per kill

Or make it:

Crushing Wheels create



+20 other recipes in JEI

Ingredient	Where it comes from
Diamond Horse Armor	Chest loot in the Ruined Citadel (The Moon), 4.6% per chest Chest loot in the Ruined Citadel (The Moon), 7.6% per chest Chest loot in the Cursed Pyramid (Mars), 6.3% per chest Chest loot in the Desert Occupied Village (Mars), 10.6% per chest

#123 Amethyst

Amethyst grows on giant crabs and inside geodes!

After **#117 One Small Step**

- Collect 16× **Amethyst Shard**

Item	Where it comes from
Amethyst Shard	Sift Moon Sand with Advanced Brass Mesh: **10%** Sift Gravel with Advanced Brass Mesh: **1%** Amethyst Crab (The Moon) drops it, 100% per kill Glacio: found in world features (Amethyst Cluster) Chest loot in the Monastery (Mars), 10.5% per chest

#124 Ice Shards

After **#117 One Small Step**

- Collect 16× **Ice Shard**

Item	Where it comes from
Ice Shard	Sift Mars Sand with Advanced Brass Mesh: **20%** Sift Conglomerate with Advanced Brass Mesh: **10%** Sift Moon Sand with Advanced Brass Mesh: **10%** The Moon: Moon Ice Shard Ore → Ice Shard (Y -32 to 32) The Moon: found in world features (Moon Ice Shard Ore) Mars: Mars Ice Shard Ore → Ice Shard (Y -32 to 32)

#125 Eyes of Ender

After #71 Blaze Powder, #119 Ender Pearls

- Collect 12× **Eye of Ender**

Crafting Table (shapeless) vanilla



Ingredient	Where it comes from
Ender Pearl	Made in #119 Ender Pearls Sift Crushed end stone with Advanced Brass Mesh: **10%** Enderman (The Moon) drops it, 100% per kill
Blaze Powder	Made in #71 Blaze Powder

#126 Void Core

Dropped by Ender Golems in the citadel

After #118 Ruined Citadel

- Collect 2× **Void Core**

> Needed for spawners, the soul catcher and enderite.

The Ender Golem's loot table drops a void core every kill. Golems are placed by the citadel's own code, so their spawn does not show up in the structure files.

Item	Where it comes from
Void Core	Ender Golem drops it (100% per kill; where it spawns is not in the data files)

#127 Ender Guardian

Optional boss

After #118 Ruined Citadel

- Defeat **Ender Guardian**

Drops (from the loot table)

Gauntlet of Guard
1 · 100%

Music Disc
0–1 · 10%

#128 The Stronghold

The only stronghold is on the Moon

After #125 Eyes of Ender

- Find the **Stronghold** on The Moon

#129 Moon Sand

Moon sand can be seived too!

After #120 Moon Cobblestone

- Collect 16× **Moon Sand**

Millstone Quicreate



Ingredient	Where it comes from
Moon Cobblestone	Made in #120 Moon Cobblestone Mechanical Extruder with Moon Cobblestone below (Lava + Water) The Moon: mine the Moon Stone terrain

Item	Also from
Moon Sand	The Moon: mine the Moon Sand terrain The Moon: found in world features (Moon Sand) Chest loot in the Lunar Tower (The Moon), 89.2% per chest Chest loot in the Lunarian Village (The Moon), 89.2% per chest Chest loot in the Moon Dungeon (The Moon), 79.4% per chest Chest loot in the Moon Dungeon (The Moon), 66.8% per chest

#130 Desh Plates

Press desh ingots

After #121 Desh

- Collect 16× **Desh Plate**

Mechanical Press ad astra



Compressor Quicreate



+1 other recipe in JEI

Ingredient	Where it comes from
Desh Ingot	Made in #121 Desh Chest loot in the Lunar Tower (The Moon), 86.7% per chest Chest loot in the Lunar Tower (The Moon), 20.9% per chest

#131 Overcharged Diamond

After #122 Diamonds, #62 Energizer

- Collect 4× **Overcharged Diamond**

Energiser 10000 energy create-new-age



Ingredient	Where it comes from
Diamond	Made in #122 Diamonds Sift Venus Sand with Advanced Brass Mesh: **5%** Sift Mars Sand with Advanced Brass Mesh: **2.5%**

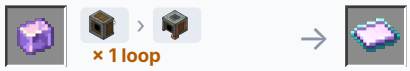
#132 Amethyst Wafer

Polish, cut and energise amethyst

After #62 Energizer, #123 Amethyst

- Collect 2× **Amethyst Wafer**

Sequenced Assembly Quicreate



Ingredient	Where it comes from
Polished Amethyst	Sand Paper (hold and use): Amethyst Shard

#133 Etrium

Energise ice shards

After #62 Energizer, #124 Ice Shards

- Collect 4× **Etrium Ingot**

Crafting Table Quicreate



Ingredient	Where it comes from
Etrium Nugget	Energiser (10000 energy): Ice Shard

#134 The End

After #128 The Stronghold

- Travel to **The End**

#135 Advanced Brass Mesh

Desh and steel around a brass mesh

After #130 Desh Plates, #99 Steel Plates, #38 Brass Mesh

- Collect 1× **Advanced Brass Mesh**

> Sift moon sand, sand, gravel and dust for diamonds, emeralds, lapis and more.

Crafting Table pack script



Ingredient	Where it comes from
Desh Plate	Made in #130 Desh Plates
Steel Plate	Made in #99 Steel Plates
Brass Mesh	Made in #38 Brass Mesh

#136 Fluxuated Magnetite

After #89 Layered Magnet, #131 Overcharged Diamond

- Collect 2× **Fluxuated Magnetite**

Crafting Table pack script



Ingredient	Where it comes from
Layered Magnet	Made in #89 Layered Magnet
Overcharged Iron Sheet	Energiser (1000 energy): any Iron Plate
Overcharged Diamond	Made in #131 Overcharged Diamond

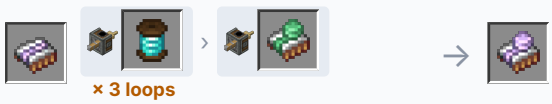
#137 Amethyst Circuit

Overcharged diamond wire and emerald circuits

After #91 Emerald Circuit, #131 Overcharged Diamond, #132 Amethyst Wafer

- Collect 4× **Amethyst Circuit**

Sequenced Assembly Quicreate



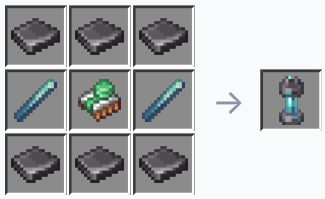
Ingredient	Where it comes from
Incomplete Amethyst Circuit	Deployer: Blank Circuit + Amethyst Wafer
Overcharged Diamond Wire	Sequenced Assembly: Overcharged Diamond → 2
Emerald Circuit	Made in #91 Emerald Circuit

#138 Etrionic Capacitor

After #91 Emerald Circuit, #133 Etrium

- Collect 2× **Etrionic Capacitor**

Crafting Table pack script



Ingredient	Where it comes from
Steel Plate	Made in #99 Steel Plates
Etrium Rod	Crafting Table: 4× Etrium Nugget + Steel Rod → 2
Emerald Circuit	Made in #91 Emerald Circuit

#139 Etrionic Core

After #99 Steel Plates, #133 Etrium

- Collect 1× **Etrionic Core**

Crafting Table Quicreate



Ingredient	Where it comes from
Steel Plate	Made in #99 Steel Plates
Etrium Rod	Crafting Table: 4× Etrium Nugget + Steel Rod → 2

#140 Free the End

The soft ending

After #134 The End

- Defeat **Ender Dragon**

> You beat the pack! Keep going for the planets and the creative ending.

The End is reached through the stronghold's portal on the Moon. This is the pack's soft ending.

Drops: Ender Dragon has no loot table in the files (its drops are handled by the game code).

#141 Desh Tanks

After #130 Desh Plates, #105 Steel Tanks

- Collect 4× Desh Tank

Crafting Table pack script



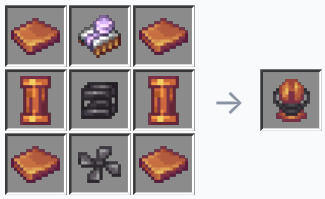
Ingredient	Where it comes from
Desh Plate	Made in #130 Desh Plates
Gas Tank	Made in #102 Gas Tanks Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest
Steel Tank	Made in #105 Steel Tanks

#142 Desh Engine

After #130 Desh Plates, #106 Rocket Fan, #137 Amethyst Circuit

- Collect 1× Desh Engine

Crafting Table pack script



Ingredient	Where it comes from
Desh Plate	Made in #130 Desh Plates
Amethyst Circuit	Made in #137 Amethyst Circuit
Desh Fluid Pipe	Crafting Table: 6× Desh Plate + 3× Glass → 16
Engine Frame	Crafting Table: 8× Steel Rod + Steel Plate
Steel Rocket Propeller	Made in #106 Rocket Fan

#143 Smart Crafter

Optional automation

After #137 Amethyst Circuit

- Collect 1× **Smart Crafter**

Mechanical Crafters Quicreate



Ingredient	Where it comes from
Cogwheel	Made in #17 Shafts & Cogwheels
Amethyst Circuit	Made in #137 Amethyst Circuit
Mechanical Crafter	Made in #73 Mechanical Crafters

#144 Chunk Loader

Optional automation

After #137 Amethyst Circuit

- Collect 1× **Empty Andesite Chunk Loader**

Mechanical Crafters Quicreate



Ingredient	Where it comes from
any Framed Glass	Stonecutter: any Colorless Gla
Iron Bars	Crafting Table: 6× Iron Ingot → 16
Amethyst Circuit	Made in #137 Amethyst Circuit
Andesite Casing	Made in #16 Andesite Casing
Shaft	Made in #17 Shafts & Cogwheels

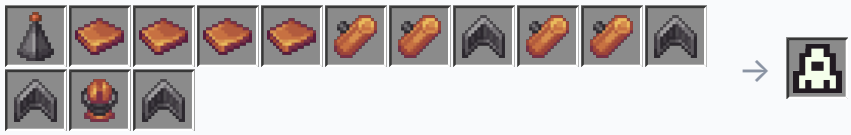
#145 Tier 2 Rocket

Next stop: Mars

After **#142 Desh Engine**, **#141 Desh Tanks**

- Collect 1× **Tier 2 Rocket**

NASA Workbench Quicreate



Ingredient	Where it comes from
Rocket Nose Cone	Made in **#100 Nose Cone & Fins**
Desh Plate	Made in **#130 Desh Plates**
Desh Tank	Made in **#141 Desh Tanks**
Rocket Fin	Made in **#100 Nose Cone & Fins**
Desh Engine	Made in **#142 Desh Engine**



CHAPTER 3: MARS · STEPS #146–#164

Mars

Ostrum only comes from Mars (ore, and sifting Mars sand or conglomerate with the advanced brass mesh). Ancient debris is in the Mars rock: the worldgen places it at Y 90 to 130 and the conglomerate boulders can be sifted for it. Beat the Ancient Remnant in the Cursed Pyramid for ancient metal, which leads to the ancient metal circuit, the ostrum engine and the netherite space suit you need for the next two planets.

Mars: what is here

Biomes: Martian Wastelands, Martian Canyon Creek, Martian Polar Caps

Terrain: Mars Stone Mars Sand Snow Block

Ore / deposit	Drops	Height	Veins / chunk	Vein size
Mars Diamond Ore	Diamond	-80 to 80	5	7
Mars Ice Shard Ore	Ice Shard	-32 to 32	8	10
Mars Iron Ore	Raw Iron	-24 to 56	10	11
Mars Ostrum Ore	Raw Ostrum	-80 to 80	8	8
Ancient Debris	itself	90 to 130	200	2

Surface features (from the worldgen files):

- **mars rock**: Conglomerate

Mob (natural spawns)	Drops per kill
Martian Raptor	no loot table in the files

Structure (biomes)	Mobs inside	Chest loot highlights
Mars Temple Martian Wastelands, Martian Canyon Creek	–	Mars Sand 67.6% Emerald 56.1% Ice Shard 22.9% String 22.9% Diamond 12.1% Redstone Dust 12.1% Ice 67.6% Book 41.4%
Cursed Pyramid Martian Wastelands	–	String 59% Sand 59% Emerald 18.2% Diamond 6.3% Bone 70.9% Rotten Flesh 70.9% Gunpowder 59% Spider Eye 29%
Desert Occupied Village Martian Wastelands	Koboleton, Wadjet, Armor Stand, Camel, Raw Chicken, Iron Golem, Item Frame, Villager	Emerald 33% Ancient Metal Ingot 29.4% Diamond 9.9% Stick 92.6% Bread 87.5% Wheat Seeds 61.5% Feather 59.6% Flint 59.6%
Desert Site Martian Wastelands	Koboleton, Wadjet	String 59% Sand 59% Ancient Metal Ingot 29.4% Emerald 29.4% Diamond 8.5% Bone 70.9% Rotten Flesh 70.9% Gunpowder 59%
Monastery Martian Wastelands, Martian Canyon Creek, Martian Polar Caps	Armor Stand, Item Frame, Bluff, Tongbi, the Sculptor	Amethyst Shard 10.5% Emerald 10.5% Stone 61.7% Diorite 44% Cooked Mutton 44% Book 44% Paper 44% Apple 44%
Mars Arch Martian Canyon Creek	–	–

Boss	Found in	Step
Ancient Remnant	Cursed Pyramid	#151 Ancient Remnant

#146 The Red Planet

Land on Mars

After #145 Tier 2 Rocket

- Travel to **Mars**

#147 Martian Sifting

Mill mars cobblestone, sift with an advanced mesh

After #146 The Red Planet, #135 Advanced Brass Mesh

- Collect 16× **Mars Sand**

Item	Where it comes from
Mars Sand	Sift Conglomerate with Advanced Brass Mesh: **20%** Mars: mine the Mars Sand terrain Chest loot in the Mars Temple (Mars), 67.6% per chest

Or make it:

Millstone Quicreate



Ingredient	Where it comes from
Mars Cobblestone	Mechanical Extruder with Mars Cobblestone below (Lava + Water) Mars: mine the Mars Stone terrain

#148 Ostrum

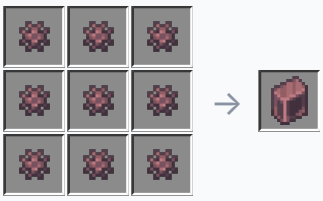
After #146 The Red Planet

- Collect 16× **Ostrum Ingot**

Furnace ad astra



Crafting Table ad astra



+6 other recipes in JEI

Ingredient	Where it comes from
Raw Ostrum	Crafting Table (shapeless): Raw Ostrum Block → 9 Mars: Mars Ostrum Ore → Raw Ostrum (Y -80 to 80) Chest loot in the Mars Temple (Mars), 12.1% per chest

#149 Ancient Debris

Buried in conglomerate

After #146 The Red Planet

- Collect 4× **Ancient Debris**

Item	Where it comes from
Ancient Debris	Sift Conglomerate with Advanced Brass Mesh: **0.5%** Fishing with the Ostrum Fishing Rod: 1% per catch Mars: Ancient Debris (Y 90 to 130)

#150 Cursed Pyramid

After #146 The Red Planet

- Find the **Cursed Pyramid** on Mars

#151 Ancient Remnant

Boss

After #150 Cursed Pyramid

- Defeat **Ancient Remnant**

Drops (from the loot table)

Sandstorm in a Bottle	Remnant's Skull	Block of Ancient Metal	Music Disc
1 · 100%	1 · 100%	1 · 100%	0–1 · 10%

#152 Ostrum Plates

After #148 Ostrum

- Collect 16× **Ostrum Plate**

Mechanical Press ad astra



Compressor Quicreate



+1 other recipe in JEI

Ingredient	Where it comes from
Ostrum Ingot	Made in #148 Ostrum

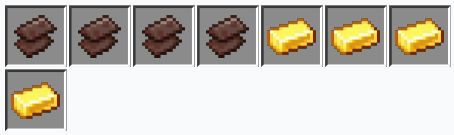
#153 Netherite

After #149 Ancient Debris

- Collect 2× **Netherite Ingot**

> The upgrade template is 8 netherrack around a diamond.

Crafting Table (shapeless) vanilla



Crafting Table (shapeless) vanilla



Ingredient	Where it comes from
Netherite Scrap	Furnace: Ancient Debris
Gold Ingot	Open a Medium Kelpy Ball (fishing): 13.1% per ball Crushing Wheels: Golden Horse Armor → 2 Chest loot in the Moon Dungeon (The Moon), 22.6% per chest Chest loot in the Ruined Citadel (The Moon), 54% per chest

#154 Ancient Metal

After #151 Ancient Remnant

- Collect 4× **Ancient Metal Ingot**

Mechanical Mixer + Basin Superheated — Quicreate



Crafting Table — L Enders Cataclysm



+2 other recipes in JEI

Ingredient	Where it comes from
Ancient Metal Nugget	Furnace: Khopesh Koboleton in the Desert Occupied Village (Mars), 100% per kill Koboleton in the Desert Site (Mars), 100% per kill
Platinum	Covered later in #170 Platinum

Item	Also from
Ancient Metal Ingot	Koboleton in the Desert Occupied Village (Mars), 2.5% per kill Wadjet in the Desert Occupied Village (Mars), 100% per kill Wadjet in the Desert Site (Mars), 100% per kill Koboleton in the Desert Site (Mars), 2.5% per kill Chest loot in the Desert Occupied Village (Mars), 29.4% per chest Chest loot in the Desert Site (Mars), 29.4% per chest

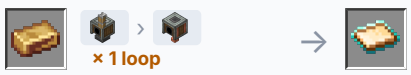
#155 Ancient Metal Wafer

Press and energise ancient metal

After #154 Ancient Metal

- Collect 2× **Ancient Metal Wafer**

Sequenced Assembly — Quicreate



Ingredient	Where it comes from
Ancient Metal Ingot	Made in #154 Ancient Metal Koboleton in the Desert Occupied Village (Mars), 2.5% per kill Wadjet in the Desert Occupied Village (Mars), 100% per kill

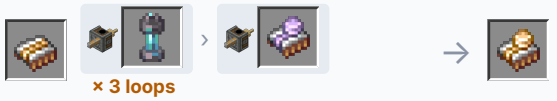
#156 Ancient Metal Circuit

Etrionic capacitors and amethyst circuits

After #155 Ancient Metal Wafer, #138 Etrionic Capacitor, #137 Amethyst Circuit

- Collect 4× **Ancient Metal Circuit**

Sequenced Assembly Quicreate



Ingredient	Where it comes from
Incomplete Ancient Metal Circuit	Deployer: Blank Circuit + Ancient Metal Wafer
Etrionic Capacitor	Made in #138 Etrionic Capacitor
Amethyst Circuit	Made in #137 Amethyst Circuit

#157 Reinforced Motor

After #137 Amethyst Circuit, #136 Fluxuated Magnetite

- Collect 1× **Reinforced Motor**

Mechanical Crafters Quicreate



Ingredient	Where it comes from
Overcharged Iron Sheet	Energiser (1000 energy): any Iron Plate
Overcharged Diamond	Made in #131 Overcharged Diamond
Sturdy Casing	Use item on block (hand or Deployer): Brass Casing + Sturdy Sheet
Amethyst Circuit	Made in #137 Amethyst Circuit
Fluxuated Magnetite	Made in #136 Fluxuated Magnetite
Shaft	Made in #17 Shafts & Cogwheels

#158 Soul Catcher

Optional automation

After **#156 Ancient Metal Circuit**, **#126 Void Core**

- Collect 1× **Empty Soul Catcher**

Crafting Table Quicreate



Ingredient	Where it comes from
Sturdy Casing	Use item on block (hand or Deployer): Brass Casing + Sturdy Sheet
Shulker Shell	Sift Crushed end stone with Advanced Brass Mesh: **1%** Shulker drops it (50% per kill; where it spawns is not in the data files)
Ancient Metal Circuit	Made in **#156 Ancient Metal Circuit**
Void Core	Made in **#126 Void Core** Ender Golem drops it (100% per kill; where it spawns is not in the data files)

#159 Netherite Space Suit

Heat protection for Venus and Mercury

After #153 Netherite, #152 Ostrum Plates, #156 Ancient Metal Circuit

- Collect 1× **Netherite Space Helmet**
- Collect 1× **Netherite Space Suit**
- Collect 1× **Netherite Space Pants**
- Collect 1× **Netherite Space Boots**

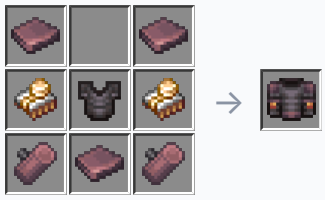
> Cataclysm cursium armor also counts.

Crafting Table — ad astra



Ingredient	Where it comes from
Ostrum Plate	Made in #152 Ostrum Plates
Netherite Helmet	Smithing Table: Netherite Upgrade Smithing Template + Diamond Helmet + Netherite Ingot
Glass	Encased Fan + Water (washing): any Black Stained Glass

Crafting Table — Quicreate



Ingredient	Where it comes from
Ostrum Plate	Made in #152 Ostrum Plates
Ancient Metal Circuit	Made in #156 Ancient Metal Circuit
Netherite Chestplate	Smithing Table: Netherite Upgrade Smithing Template + Diamond Chestplate + Netherite Ingot
Ostrum Tank	Covered later in #160 Ostrum Tanks

Crafting Table — ad astra



Ingredient	Where it comes from
Ostrum Plate	Made in #152 Ostrum Plates
Netherite Leggings	Smithing Table: Netherite Upgrade Smithing Template + Diamond Leggings + Netherite Ingot
Desh Plate	Made in #130 Desh Plates

Crafting Table — ad astra



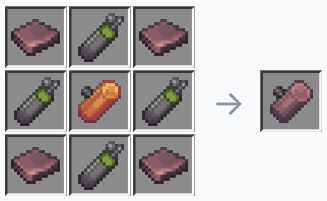
Ingredient	Where it comes from
Netherite Boots	Smithing Table: Netherite Upgrade Smithing Template + Diamond Boots + Netherite Ingot
Desh Plate	Made in #130 Desh Plates
Ostrum Plate	Made in #152 Ostrum Plates

#160 Ostrum Tanks

After #152 Ostrum Plates, #141 Desh Tanks

- Collect 4× **Ostrum Tank**

Crafting Table — pack script



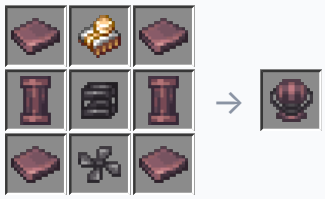
Ingredient	Where it comes from
Ostrum Plate	Made in #152 Ostrum Plates
Gas Tank	Made in #102 Gas Tanks Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest
Desh Tank	Made in #141 Desh Tanks

#161 Ostrum Engine

After #156 Ancient Metal Circuit, #152 Ostrum Plates

- Collect 1× **Ostrum Engine**

Crafting Table pack script



Ingredient	Where it comes from
Ostrum Plate	Made in #152 Ostrum Plates
Ancient Metal Circuit	Made in #156 Ancient Metal Circuit
Ostrum Fluid Pipe	Crafting Table: 6× Ostrum Plate + 3× Glass → 16
Engine Frame	Crafting Table: 8× Steel Rod + Steel Plate
Steel Rocket Propeller	Made in #106 Rocket Fan

#162 Brass Chunk Loader

Optional automation

After #156 Ancient Metal Circuit, #144 Chunk Loader

- Collect 1× **Empty Brass Chunk Loader**

Mechanical Crafters Quicreate



Ingredient	Where it comes from
Tinted Glass	Crafting Table: 4× Amethyst Shard + Glass → 2
Empty Andesite Chunk Loader	Made in #144 Chunk Loader
Ancient Metal Circuit	Made in #156 Ancient Metal Circuit
Sturdy Casing	Use item on block (hand or Deployer): Brass Casing + Sturdy Sheet
Shaft	Made in #17 Shafts & Cogwheels

#163 Mechanical Spawner

Optional automation

After #156 Ancient Metal Circuit, #126 Void Core

- Collect 1× **Mechanical Spawner**

Mechanical Crafters Quicreate



Ingredient	Where it comes from
Sturdy Casing	Use item on block (hand or Deployer): Brass Casing + Sturdy Sheet
Ancient Metal Circuit	Made in #156 Ancient Metal Circuit
Iron Bars	Crafting Table: 6× Iron Ingot → 16
Void Core	Made in #126 Void Core Ender Golem drops it (100% per kill; where it spawns is not in the data files)

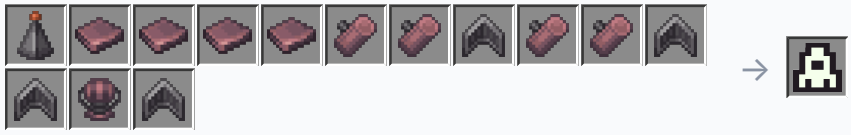
#164 Tier 3 Rocket

Next stop: Venus and Mercury

After #161 Ostrum Engine, #160 Ostrum Tanks

- Collect 1× **Tier 3 Rocket**

NASA Workbench Quicreate



Ingredient	Where it comes from
Rocket Nose Cone	Made in #100 Nose Cone & Fins
Ostrum Plate	Made in #152 Ostrum Plates
Ostrum Tank	Made in #160 Ostrum Tanks
Rocket Fin	Made in #100 Nose Cone & Fins
Ostrum Engine	Made in #161 Ostrum Engine



CHAPTER 4: VENUS & MERCURY · STEPS #165–#184

Venus and Mercury

Both planets need heat protection: the quest book asks for the netherite space suit before you land (cursium armour also counts, see **#159 Netherite Space Suit**). Venus has calorite, the Ancient Factory (Harbinger) and the new Fungal Forest, the only biome with Umvuthana Groves. Mercury is the only planet with platinum ore, the Burning Arena (Ignis) and the Soul Black Smith (Netherite Monstrosity).

Venus: what is here

Biomes: Venus Wastelands, Infernal Venus Barrens, Sandstone Dunes, Fungal Forest

Terrain: Venus Stone Venus Sand Venus Sandstone Basalt Gravel Blackstone

Ore / deposit	Drops	Height	Veins / chunk	Vein size
Venus Calorite Ore	Raw Calorite	-80 to 80	8	8
Venus Coal Ore	Coal	-80 to 192	20	17
Venus Diamond Ore	Diamond	-80 to 80	5	9
Venus Gold Ore	Raw Gold	-64 to 32	4	10
Magma Block	itself	27 to 36	4	33

Surface features (from the worldgen files):

- **venus sandstone boulder**: Venus Sandstone Venus Stone
- **venus strophar mushroom large**: Strophar Stem Strophar Cap Shroomlight
- **venus strophar mushroom**: Strophar Stem Strophar Cap Shroomlight
- **venus strophar patch**: Strophar Mushroom

Mob (natural spawns)	Drops per kill
Mogler	no loot table in the files
Sulfur Creeper	no loot table in the files
Zombified Pygro	no loot table in the files
Pygro	no loot table in the files

Structure (biomes)	Mobs inside	Chest loot highlights
Ancient Factory Infernal Venus Barrens, Sandstone Dunes, Fungal Forest	The Harbinger, The Prowler, The Watcher, Minecart with Hopper	–
Umvuthana Grove Fungal Forest	–	–

Mercury: what is here

Biomes: Mercury Deltas, Shattered Rift

Terrain: Mercury Stone Blackstone Basalt Gravel

Ore / deposit	Drops	Height	Veins / chunk	Vein size
Magma Block	itself	27 to 36	4	33
Mercury Iron Ore	Raw Iron	-80 to 192	20	8
Platinum Ore	Raw Platinum	0 to 30	6	4

Surface features (from the worldgen files):

- **mercury fissure sparse**: Lava Blackstone Basalt Magma Block
- **mercury lone spike**: Mercury Stone Mercury Cobblestone Magma Block

- **mercury fissure**: Lava Blackstone Basalt Magma Block
- **mercury spike cluster**: Mercury Stone Mercury Cobblestone Magma Block
- **mercury monolith ring**: Mercury Stone Bricks Cracked Mercury Stone Bricks Chiseled Mercury Stone Bricks

Mob (natural spawns)	Drops per kill
Magma Cube	Magma Cream 25%, Pearlescent Froglight 25%, Verdant Froglight 25%, Ochre Froglight 25%
Ignited Berserker	Dying Ember 100%
Wither Skeleton	Coal 100%, Bone 100%, Wither Skeleton Skull 2.5%

Structure (biomes)	Mobs inside	Chest loot highlights
Burning Arena Mercury Deltas, Shattered Rift	–	–
Soul Black Smith Mercury Deltas, Shattered Rift	Netherite Ministrosity, Netherite Monstrosity	–

Boss	Found in	Step
The Harbinger	Ancient Factory	#169 The Harbinger
Umvuthi, the Sunbird (optional)	–	#174 Umvuthi
Ignis	Burning Arena	#177 Ignis
Netherite Monstrosity	Soul Black Smith	#178 Netherite Monstrosity

#165 Acid Rain

Land on Venus

After #164 Tier 3 Rocket, #159 Netherite Space Suit

- Travel to **Venus**

#166 Closest to the Sun

Land on Mercury

After #164 Tier 3 Rocket, #159 Netherite Space Suit

- Travel to **Mercury**

#167 Calorite

After #165 Acid Rain

- Collect 16× **Calorite Ingot**

Furnace ad astra



Crafting Table ad astra



+6 other recipes in JEI

Ingredient	Where it comes from
Raw Calorite	Crafting Table (shapeless): Raw Calorite Block → 9 Venus: Venus Calorite Ore → Raw Calorite (Y -80 to 80)

#168 Ancient Factory

After #165 Acid Rain

- Find the **Ancient Factory** on Venus

#169 The Harbinger

Boss

After #168 Ancient Factory

- Defeat **The Harbinger**

Drops (from the loot table)

Block of Witherite	Music Disc
1 · 100%	0–1 · 10%

#170 Platinum

Only found on Mercury

After #166 Closest to the Sun

- Collect 16× **Platinum**

Furnace cards



Crafting Table (shapeless) cards



+1 other recipe in JEI

Ingredient	Where it comes from
Raw Platinum	Mercury: Platinum Ore → Raw Platinum (Y 0 to 30)

#171 Burning Arena

After #166 Closest to the Sun

- Find the **Burning Arena** on Mercury

#172 Soul Black Smith

After #166 Closest to the Sun

- Find the **Soul Black Smith** on Mercury

#173 Calorite Plates

After #167 Calorite

- Collect 24× **Calorite Plate**

Mechanical Press ad astra



Compressor Quicreate



+1 other recipe in JEI

Ingredient	Where it comes from
Calorite Ingot	Made in #167 Calorite

#174 Umvuthi

Optional boss in the Venus wastelands

After #165 Acid Rain

- Defeat **Umvuthi, the Sunbird**

The datapack places Umvuthana Groves only in the Venus **Fungal Forest** biome (`mowziesmobs/worldgen/structure/umvuthana_grove.json`), so look there rather than in the wastelands.

Drops (from the loot table)

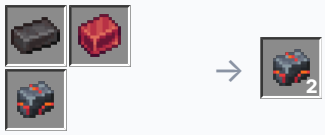
Sol Visage	1 · 100%
Feather	0–4 · 100%

#175 Witherite

After #169 The Harbinger

- Collect 4× **Witherite Ingot**

Mechanical Mixer + Basin Superheated — Quicreate



Crafting Table (shapeless) — L Enders Cataclysm



+1 other recipe in JEI

Ingredient	Where it comes from
Netherite Ingot	Made in #153 Netherite
Calorite Ingot	Made in #167 Calorite

#176 Burning Ashes

Craft from dying embers dropped by Ignited Berserkers

After #166 Closest to the Sun

- Collect 4× **Burning Ashes**

Crafting Table — L Enders Cataclysm



Ingredient	Where it comes from
Dying Ember	Ignited Berserker (Mercury) drops it, 100% per kill

Item	Also from
Burning Ashes	Ignited Revenant drops it (100% per kill; where it spawns is not in the data files)

#177 Ignis

Boss

After #171 Burning Arena

- Defeat **Ignis**

Drops (from the loot table)

Ignitium Ingot	3 · 100%
Music Disc	0–1 · 10%

#178 Netherite Monstrosity

Boss

After #172 Soul Black Smith

- Defeat **Netherite Monstrosity**

Drops (from the loot table)

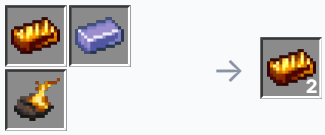
Infernal Forge	Monstrous Horn	Lava Power Cell	Music Disc
1 · 100%	1 · 100%	16–24 · 100%	0–1 · 10%

#179 Ignitium

After #177 Ignis

- Collect 4× **Ignitium Ingot**

Mechanical Mixer + Basin Superheated — Quicreate



Crafting Table (shapeless) — L Enders Cataclysm



+1 other recipe in JEI

Ingredient	Where it comes from
Platinum	Made in #170 Platinum
Burning Ashes	Made in #176 Burning Ashes Ignited Revenant drops it (100% per kill; where it spawns is not in the data files)

Item	Also from
Ignitium Ingot	Ignis boss drop (Mercury), 100%

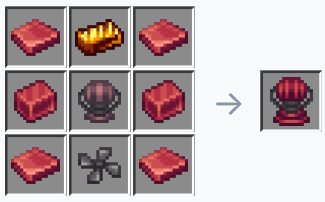
#180 Calorite Engine

Ignitium drives the last engine

After #173 Calorite Plates, #179 Ignitium, #161 Ostrum Engine

- Collect 1× **Calorite Engine**

Crafting Table — pack script



Ingredient	Where it comes from
Calorite Plate	Made in #173 Calorite Plates
Ignitium Ingot	Made in #179 Ignitium Ignis boss drop (Mercury), 100%
Calorite Ingot	Made in #167 Calorite
Ostrum Engine	Made in #161 Ostrum Engine
Steel Rocket Propeller	Made in #106 Rocket Fan



#181 Calorite Tanks

After #173 Calorite Plates, #160 Ostrum Tanks

- Collect 4×  **Calorite Tank**

Crafting Table pack script



Ingredient	Where it comes from
 **Calorite Plate**	Made in #173 Calorite Plates
 **Gas Tank**	Made in #102 Gas Tanks Chest loot in the Lunar Tower (The Moon), 18.2% per chest Chest loot in the Lunarian Village (The Moon), 18.2% per chest
 **Ostrum Tank**	Made in #160 Ostrum Tanks

#182 Jet Suit

Ignitium armor also counts

After #173 Calorite Plates, #181 Calorite Tanks, #138 Etrionic Capacitor, #159 Netherite Space Suit

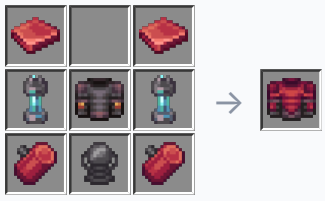
- Collect 1× **Jet Suit Helmet**
- Collect 1× **Jet Suit**
- Collect 1× **Jet Suit Pants**
- Collect 1× **Jet Suit Boots**

Crafting Table ad astra



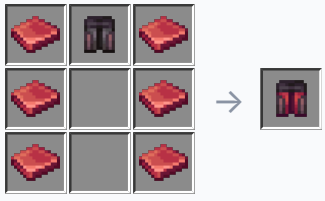
Ingredient	Where it comes from
Calorite Plate	Made in #173 Calorite Plates
Netherite Space Helmet	Made in #159 Netherite Space Suit
Tinted Glass	Crafting Table: 4× Amethyst Shard + Glass → 2

Crafting Table Quicreate



Ingredient	Where it comes from
Calorite Plate	Made in #173 Calorite Plates
Etrionic Capacitor	Made in #138 Etrionic Capacitor
Netherite Space Suit	Made in #159 Netherite Space Suit
Calorite Tank	Made in #181 Calorite Tanks
Steel Engine	Made in #111 Steel Engine

Crafting Table pack script



Ingredient	Where it comes from
Calorite Plate	Made in #173 Calorite Plates
Netherite Space Pants	Made in #159 Netherite Space Suit

Crafting Table pack script



Ingredient	Where it comes from
Netherite Space Boots	Made in #159 Netherite Space Suit
Calorite Plate	Made in #173 Calorite Plates

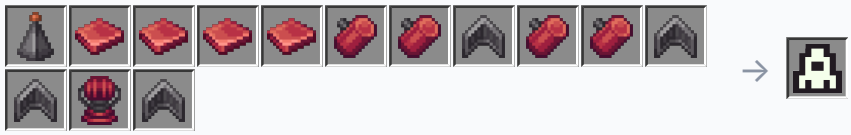
#183 Tier 4 Rocket

Next stop: Glacio

After #180 Calorite Engine, #181 Calorite Tanks

- Collect 1× **Tier 4 Rocket**

NASA Workbench Quicreate



Ingredient	Where it comes from
Rocket Nose Cone	Made in #100 Nose Cone & Fins
Calorite Plate	Made in #173 Calorite Plates
Calorite Tank	Made in #181 Calorite Tanks
Rocket Fin	Made in #100 Nose Cone & Fins
Calorite Engine	Made in #180 Calorite Engine



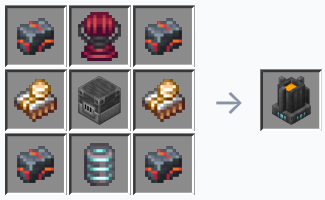
#184 Etrionic Blast Furnace

Alloying, including boss-ingot doubling

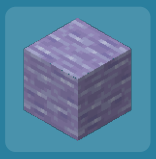
After **#175 Witherite**, **#180 Calorite Engine**, **#156 Ancient Metal Circuit**

- Collect 1× **Etrionic Blast Furnace**

Crafting Table pack script



Ingredient	Where it comes from
Witherite Ingot	Made in **#175 Witherite**
Calorite Engine	Made in **#180 Calorite Engine**
Ancient Metal Circuit	Made in **#156 Ancient Metal Circuit**
Blast Furnace	Crafting Table: 5× Iron Ingot + Furnace + 3× Smooth Stone
Etrionic Core	Made in **#139 Etrionic Core**



CHAPTER 5: GLACIO · STEPS #185–#200

Glacio

Glacio holds the last bosses. Maledictus (Frosted Prison) gives cursium, Scylla (Acropolis) gives the essence of the storm, and the Frostmaw gives the ice crystal. All three feed the creative parts in the next chapter. The Sunken City (Leviathan) and the Ferrous Wroughtnaut are optional.

Glacio: what is here

Biomes: Glacio Snowy Barrens, Glacio Ice Peaks, Glacio Helix Spires, Glacian Woods

Terrain: Glacio Stone Powder Snow Snow Block Permafrost Deepslate

Ore / deposit	Drops	Height	Veins / chunk	Vein size
Glacio Coal Ore	Coal	-80 to 192	20	17
Glacio Copper Ore	Raw Copper	-16 to 112	16	17
Glacio Ice Shard Ore	Ice Shard	-32 to 32	8	17
Glacio Iron Ore	Raw Iron	-24 to 56	10	11
Glacio Lapis Ore	Lapis Lazuli	-32 to 32	2	9
Deepslate Coal Ore	Coal	-80 to 192	20	17
Deepslate Copper Ore	Raw Copper	-80 to 112	16	17
Deepslate Iron Ore	Raw Iron	-80 to 20	10	12
Deepslate Lapis Lazuli Ore	Lapis Lazuli	-80 to 10	2	9

Surface features (from the worldgen files):

- **glacio helix spire**: Packed Ice Blue Ice Snow Block

Mob (natural spawns)	Drops per kill
Glacian Ram	no loot table in the files
Elokosa	no loot table in the files

Structure (biomes)	Mobs inside	Chest loot highlights
Abandoned Spire Glacio Ice Peaks, Glacio Helix Spires	Aptrgangr, Draugr, Elite Draugr, Royal Draugr	Black Steel Ingot 35.2% Emerald 18.8% Diamond 9.7% Leather 54% Iron Ingot 47.1% Gold Ingot 35.2% Ring of Grudged 25% Berserker Soul Amulet 25%
Abandoned Temple Glacio Ice Peaks, Glacio Helix Spires	Aptrgangr, Draugr, Elite Draugr, Royal Draugr	Black Steel Ingot 35.2% Emerald 18.8% Diamond 9.7% Leather 54% Iron Ingot 47.1% Gold Ingot 35.2% Ring of Grudged 25% Berserker Soul Amulet 25%
Abandoned Village Glacio Ice Peaks, Glacio Helix Spires	Draugr, Elite Draugr, Royal Draugr	Black Steel Ingot 35.2% Emerald 18.8% Diamond 9.7% Leather 54% Iron Ingot 47.1% Gold Ingot 35.2% Ring of Grudged 25% Berserker Soul Amulet 25%
Acropolis Glacio Ice Peaks, Glacio Helix Spires	Cindaria, Clawdian, Drowned Host, Hippocamtus, Scylla, Symbiocto, Urchinkin, Trident	Diamond 15.2% Emerald 15.2% Book 50.5% Golden Apple 50.5% Gold Ingot 40.3% Scute 33.4% Copper Ingot 28.6% Nautilus Shell 20.7%
Frosted Prison Glacio Ice Peaks, Glacio Helix Spires	Aptrgangr, Draugr, Elite Draugr, Royal Draugr	String 57.8% Black Steel Ingot 42.2% Redstone Dust 27.4% Diamond 23.3% Emerald 9.9% Bone 57.8% Gunpowder 57.8% Rotten Flesh 57.8%
Sunken City Glacio Snowy Barrens, Glacian Woods	–	Iron Nugget 96.8% Emerald 75.2% String 57.8% Gold Nugget 37.3% Redstone Dust 27.4% Bamboo 14.5% Diamond 14.1% Empty Map 100%

Frostmaw Spawn Glacio Snowy Barrens, Glacio Ice Peaks, Glacio Helix Spires, Glacian Woods	Frostmaw	–
Wrought Chamber Glacio Snowy Barrens, Glacio Ice Peaks, Glacio Helix Spires, Glacian Woods	–	–

Boss	Found in	Step
Frostmaw	–	#190 Frostmaw
Ferrous Wroughtnaut (optional)	–	#191 Ferrous Wroughtnaut
Maledictus	Frosted Prison	#193 Maledictus
The Leviathan (optional)	Sunken City	#194 The Leviathan
Scylla	Acropolis	#195 Scylla



#185 The Frozen World

Land on Glacio

After #183 Tier 4 Rocket

- Travel to **Glacio**



#186 Frosted Prison

After #185 The Frozen World

- Find the **Frosted Prison** on Glacio



#187 Sunken City

Optional

After #185 The Frozen World

- Find the **Sunken City** on Glacio



#188 Acropolis

After #185 The Frozen World

- Find the **Acropolis** on Glacio

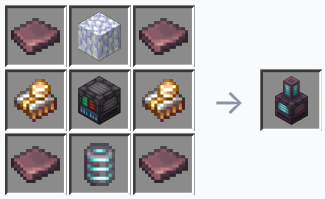
#189 Cryo Freezer

Ice into cryo fuel

After #185 The Frozen World, #156 Ancient Metal Circuit, #115 Fuel Refinery

- Collect 1× **Cryo Freezer**

Crafting Table pack script



Ingredient	Where it comes from
Ostrum Plate	Made in #152 Ostrum Plates
Permafrost	Crafting Table: 9× Ice Shard Glacio: mine the Permafrost terrain
Ancient Metal Circuit	Made in #156 Ancient Metal Circuit
Fuel Refinery	Made in #115 Fuel Refinery
Etrionic Core	Made in #139 Etrionic Core

#190 Frostmaw

Boss

After #185 The Frozen World

- Defeat **Frostmaw**

Drops (from the loot table)

Ice Crystal	1 · 100%
Music Disc	1 · 100%

#191 Ferrous Wroughtnaut

Optional boss

After #185 The Frozen World

- Defeat **Ferrous Wroughtnaut**

Drops (from the loot table)

Wrought Helm	1 · 100%
Axe of a Thousand Metals	1 · 100%

#192 Black Steel

Draugr in the prison drop it

After #186 Frosted Prison

- Collect 4× **Black Steel Ingot**

Mechanical Mixer + Basin Superheated Quicreate



Etrionic Blast Furnace (alloying) Quicreate



+2 other recipes in JEI

Ingredient	Where it comes from
Black Steel Nugget	Furnace: Black Steel Axe Elite Draugr in the Abandoned Spire (Glacio), 100% per kill Aptrgangr in the Abandoned Spire (Glacio), 100% per kill Draugr in the Abandoned Spire (Glacio), 100% per kill
Platinum	Made in #170 Platinum

Item	Also from
Black Steel Ingot	Elite Draugr in the Abandoned Spire (Glacio), 2.5% per kill Aptrgangr in the Abandoned Spire (Glacio), 100% per kill Draugr in the Abandoned Spire (Glacio), 2.5% per kill Royal Draugr in the Abandoned Spire (Glacio), 2.5% per kill Elite Draugr in the Abandoned Temple (Glacio), 2.5% per kill Aptrgangr in the Abandoned Temple (Glacio), 100% per kill

#193 Maledictus

Boss

After #186 Frosted Prison

- Defeat **Maledictus**

Drops (from the loot table)

- Music Disc — 0–1 · 10%
- Cursium Ingot — 3–4 · 100%

#194 The Leviathan

Optional boss

After #187 Sunken City

- Defeat **The Leviathan**

Drops (from the loot table)

- Tidal Claws — 1 · 100%
- Abyssal Egg — 1 · 100%
- Music Disc — 0–1 · 10%

#195 Scylla

Boss

After #188 Acropolis

- Defeat **Scylla**

Drops (from the loot table)

- Lacrima — 8–16 · 100%
- Music Disc — 0–1 · 10%
- Essence of the Storm — 3–4 · 100%

#196 Cryo Fuel

After #189 Cryo Freezer

- Collect 1× **Cryo Fuel Bucket**

Item	Where it comes from
Cryo Fuel Bucket	Fill a bucket with Cryo Fuel (Spout, or pick it up)

#197 Ice Crystal

After #190 Frostmaw

- Collect 1× **Ice Crystal**

Item	Where it comes from
Ice Crystal	Frostmaw in the Frostmaw Spawn (Glacio), 100% per kill Frostmaw boss drop (Glacio), 100%

#198 Void Steel

Black steel + ender pearl, superheated

After #192 Black Steel

- Collect 4× **Void Steel Ingot**

Mechanical Mixer + Basin Superheated Quicreate



Etrionic Blast Furnace (alloying) Quicreate



+1 other recipe in JEI

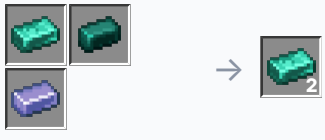
Ingredient	Where it comes from
Black Steel Ingot	Made in #192 Black Steel Elite Draugr in the Abandoned Spire (Glacio), 2.5% per kill Aptrgangr in the Abandoned Spire (Glacio), 100% per kill
Ender Pearl	Made in #119 Ender Pearls Sift Crushed end stone with Advanced Brass Mesh: **10%** Enderman (The Moon) drops it, 100% per kill

#199 Cursium

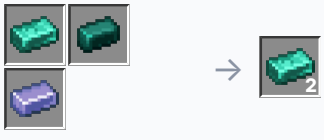
After #193 Maledictus

- Collect 4× **Cursium Ingot**

Mechanical Mixer + Basin Superheated Quicreate



Etrionic Blast Furnace (alloying) Quicreate



+1 other recipe in JEI

Ingredient	Where it comes from
Void Steel Ingot	Made in #198 Void Steel
Platinum	Made in #170 Platinum

Item	Also from
Cursium Ingot	Maledictus boss drop (Glacio), 100%



#200 Essence of the Storm

One per kill: plan your creative items

After **#195 Scylla**

- Collect 1× **Essence of the Storm**

Item	Where it comes from
Essence of the Storm	Scylla in the Acropolis (Glacio), 100% per kill Scylla boss drop (Glacio), 100%



ENDGAME: CREATIVE · STEPS #201–#213

The Creative Ending

Everything comes together: boss ingots go into the creative alloy, sixteen loops of sequenced assembly make the creative sheet, and the Mechanical Fusion Anvil turns a speed controller and a blaze cake into their creative versions. The three Quicreate swords at the end of the chapter are optional.



#201 Sturdy Sheets

After #39 Powdered Obsidian, #61 Lava

- Collect 16× **Sturdy Sheet**

Sequenced Assembly create



Ingredient	Where it comes from
Powdered Obsidian	Made in #39 Powdered Obsidian Sift Dust block with Advanced Brass Mesh: **50%** Sift Dust block with Brass Mesh: **20%**
500 mB Lava	Mechanical Mixer + Basin (Superheated): any Cobblestone



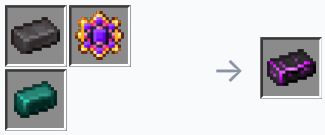
#202 Enderite

Netherite, void core and void steel

After #153 Netherite, #126 Void Core, #198 Void Steel

- Collect 4× **Enderite Ingot**

Mechanical Mixer + Basin Superheated Quicreate



Ingredient	Where it comes from
Netherite Ingot	Made in #153 Netherite
Void Core	Made in #126 Void Core Ender Golem drops it (100% per kill; where it spawns is not in the data files)
Void Steel Ingot	Made in #198 Void Steel

#203 Mechanical Fusion Anvil

Witherite, a redstone block and an anvil

After #175 Witherite

- Collect 1× **Mechanical Fusion Anvil**

Crafting Table L Enders Cataclysm



Ingredient	Where it comes from
Witherite Ingot	Made in #175 Witherite
Block of Redstone	Crafting Table: 9× Redstone Dust
Anvil	Crafting Table: 3× Block of Iron + 4× Iron Ingot

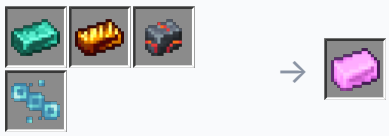
#204 Creative Alloy

Cursium, ignitium, witherite and the essence of the storm

After #199 Cursium, #179 Ignitium, #175 Witherite, #200 Essence of the Storm

- Collect 1× **Creative Alloy**

Mechanical Mixer + Basin Superheated Quicreate



Ingredient	Where it comes from
Cursium Ingot	Made in #199 Cursium Maledictus boss drop (Glacio), 100%
Ignitium Ingot	Made in #179 Ignitium Ignis boss drop (Mercury), 100%
Witherite Ingot	Made in #175 Witherite
Essence of the Storm	Made in #200 Essence of the Storm Scylla in the Acropolis (Glacio), 100% per kill Scylla boss drop (Glacio), 100%

#205 Creative Sheet

16 loops of cryo fuel, ice crystals, energy, sturdy sheets and pressing

After #204 Creative Alloy, #196 Cryo Fuel, #197 Ice Crystal, #201 Sturdy Sheets

- Collect 1× **Creative Sheet**

Sequenced Assembly Quicreate



Ingredient	Where it comes from
Creative Alloy	Made in #204 Creative Alloy
250 mB Cryo Fuel	Cryo Freezer: Blue Ice
Ice Crystal	Made in #197 Ice Crystal Frostmaw in the Frostmaw Spawn (Glacio), 100% per kill Frostmaw boss drop (Glacio), 100%
Sturdy Sheet	Made in #201 Sturdy Sheets

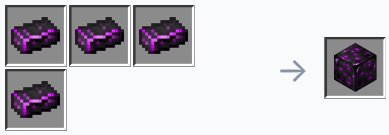
#206 Block of Enderite

Compact 4 enderite ingots

After #202 Enderite

- Collect 1× **Block of Enderite**

Mechanical Press + Basin (compacting) Superheated Quicreate



Ingredient	Where it comes from
Enderite Ingot	Made in #202 Enderite

#207 Creative Motor

Speed controller + creative alloy in the fusion anvil

After #204 Creative Alloy, #203 Mechanical Fusion Anvil

- Collect 1× **Creative Motor**

Mechanical Fusion Anvil Quicreate



Ingredient	Where it comes from
Rotation Speed Controller	Crafting Table: Precision Mechanism + Brass Casing
Creative Alloy	Made in #204 Creative Alloy

#208 Creative Blaze Cake

Blaze cake + creative alloy in the fusion anvil

After #204 Creative Alloy, #203 Mechanical Fusion Anvil

- Collect 1× **Creative Blaze Cake**

Mechanical Fusion Anvil Quicreate



Ingredient	Where it comes from
Blaze Cake	Spout: Blaze Cake Base + 250 mB Lava
Creative Alloy	Made in #204 Creative Alloy

#209 Creative Casing

A creative sheet on a block of enderite

After #205 Creative Sheet, #206 Block of Enderite

- Collect 1× **Creative Casing**

Use item on block (hand or Deployer) Quicreate



Ingredient	Where it comes from
Block of Enderite	Made in #206 Block of Enderite
Creative Sheet	Made in #205 Creative Sheet

#210 Quicreate Complete

The true ending

After #209 Creative Casing, #207 Creative Motor, #208 Creative Blaze Cake, #140 Free the End

- **Claim your victory** (tick it in the quest book)

> Every planet visited, every boss down, and creative-tier factory parts in hand. Thanks for playing!



#211 Ancient Gilded Sword

Optional Quicreate weapon

After #155 Ancient Metal Wafer, #153 Netherite, #152 Ostrum Plates

- Collect 1× **Ancient Gilded Sword**

Mechanical Crafters Quicreate



Ingredient	Where it comes from
Ostrum Plate	Made in #152 Ostrum Plates
Ancient Metal Wafer	Made in #155 Ancient Metal Wafer
Netherite Ingot	Made in #153 Netherite
Netherite Scrap	Furnace: Ancient Debris
Steel Rod	Crafting Table: 2× Steel Plate → 4



#212 Maptera Razor Sword

Optional Quicreate weapon

After #126 Void Core, #118 Ruined Citadel, #75 Overcharged Gold

- Collect 1× **Maptera Razor Sword**

> Void jaws drop from Endermapteras in the Ruined Citadel.

Mechanical Crafters Quicreate



Ingredient	Where it comes from
Void Jaw	Endermaptera in the Ruined Citadel (The Moon), 100% per kill
Diamond	Made in #122 Diamonds Sift Venus Sand with Advanced Brass Mesh: **5%** Sift Mars Sand with Advanced Brass Mesh: **2.5%**
Void Core	Made in #126 Void Core Ender Golem drops it (100% per kill; where it spawns is not in the data files)
Overcharged Golden Sheet	Energiser (2000 energy): Golden Sheet
Steel Rod	Crafting Table: 2× Steel Plate → 4

#213 Platinum Master Sword

Optional Quicreate weapon

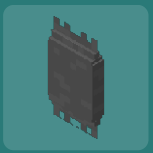
After **#170 Platinum**, **#173 Calorite Plates**, **#133 Etrium**, **#153 Netherite**

- Collect 1× **Platinum Master Sword**

Mechanical Crafters Quicreate



Ingredient	Where it comes from
Platinum	Made in **#170 Platinum**
Etrium Ingot	Made in **#133 Etrium**
Calorite Plate	Made in **#173 Calorite Plates**
Polished Amethyst	Sand Paper (hold and use): Amethyst Shard
Netherite Ingot	Made in **#153 Netherite**
Etrium Rod	Crafting Table: 4× Etrium Nugget + Steel Rod → 2



SIDE: CARDS & MOBS · STEPS #214–#220

Side chapter: Cards and Mobs

Many sifting tables drop card packs at small chances (Appendix A). Packs give cards, and the pack's CraftTweaker scripts turn mob cards into spawn eggs (Appendix D). Smelting an uncommon pack gives a nether pack, the way to wither skeletons and the Wither.

Boss	Found in	Step
Wither	–	#218 The Wither



#214 Card Packs

Rare drops from sifting

After #21 Sifter

- Collect 1× **Any card pack**

> Open packs for cards. Many mob cards craft into spawn eggs.



#215 Farm Animals

Wool, leather, eggs

After #214 Card Packs

- Collect 1× **Sheep Spawn Egg**
- Collect 1× **Chicken Spawn Egg**

> Common packs hold cow, sheep, pig and chicken cards.

Crafting Table (shapeless) pack script



Crafting Table (shapeless) pack script



+1 other recipe in JEI

Ingredient	Where it comes from
Golden Sheep Card	Crafting Table (shapeless): Platinum Base Card + Sheep Card + Gold Ingot

Crafting Table (shapeless) pack script



Ingredient	Where it comes from
Chicken Card	Deployer (held item is kept): any Common Card Pack + Shears

#216 Nether Pack

Smelt an uncommon pack

After #214 Card Packs, #9 furnace

- Collect 1× **Nether Card Pack**

Furnace pack script



Blast Furnace pack script



+1 other recipe in JEI

Ingredient	Where it comes from
Uncommon Card Pack	Sift Gravel with Brass Mesh: **1%** Sift Sand with Brass Mesh: **1%** Sift Dust block with Brass Mesh: **1%**

#217 Wither Skeletons

Skulls for a Wither

After #216 Nether Pack

- Collect 3× **Wither Skeleton Skull**

Item	Where it comes from
Wither Skeleton Skull	Wither Skeleton (Mercury) drops it, 2.5% per kill

#218 The Wither

Nether stars for beacons

After #217 Wither Skeletons

- Defeat **Wither**

Drops: Wither has no loot table in the files (its drops are handled by the game code).

#219 Flight Anchor

Creative flight near your base

After #218 The Wither, #156 Ancient Metal Circuit, #131 Overcharged Diamond

- Collect 1× **Flight Anchor**

Mechanical Crafters Quicreate



Ingredient	Where it comes from
Sturdy Casing	Use item on block (hand or Deployer): Brass Casing + Sturdy Sheet
Overcharged Diamond	Made in #131 Overcharged Diamond
Ancient Metal Circuit	Made in #156 Ancient Metal Circuit
Beacon	Crafting Table: 5× Glass + Nether Star + 3× Obsidian



#220 Villagers

From rare packs

After **#214 Card Packs**

- Collect 1× **Villager Spawn Egg**

> Trade for emeralds and enchanted books.

Crafting Table pack script



Crafting Table pack script



+1 other recipe in JEI

Ingredient	Where it comes from
Card Trader Villager Card	Deployer (held item is kept): any Rare Card Pack + Shears

Appendix A · Sifting tables

Every `createsifter:sifting` recipe in the pack. Chances are per sifted block.

Dirt

String Mesh	Beetroot Seeds 10% · Wheat Seeds 10% · Common Card Pack 0.1%
Andesite Mesh	Pumpkin Seeds 10% · Melon Seeds 10% · Red Mushroom 5% · Brown Mushroom 5% · Uncommon Card Pack 0.25% · Common Card Pack 0.5%
Zinc Mesh	Potato 10% · Carrot 10% · Rare Card Pack 0.25% · Uncommon Card Pack 0.5% · Common Card Pack 1%
String Mesh waterlogged	Oak Sapling 10% · Birch Sapling 10% · Spruce Sapling 10% · Jungle Sapling 10% · Acacia Sapling 10% · Dark Oak Sapling 10% · Bamboo 10% · Water Card Pack 0.1%

Gravel

Andesite Mesh	Coal 30% · Flint 20% · Uncommon Card Pack 0.25% · Common Card Pack 0.5%
Zinc Mesh	Copper Nugget 35% · Coal 20% · Flint 10% · Rare Card Pack 0.25% · Uncommon Card Pack 0.5% · Common Card Pack 1%
Brass Mesh	Crushed Raw Copper 12.5% · Thorium 2.5% · Nether Quartz 7.5% · Coal 10% · Flint 5% · Rare Card Pack 0.5% · Uncommon Card Pack 1% · Common Card Pack 2.5%
Advanced Brass Mesh	Raw Copper 40% · Raw Iron 40% · Nether Quartz 20% · Amethyst Shard 1% · Lapis Lazuli 10% · Thorium 10% · Ultimate Card Pack 0.1%

Sand

String Mesh	Iron Nugget 25% · Sugar Cane 2.5% · Cactus 2.5% · Common Card Pack 0.1%
Andesite Mesh	Iron Nugget 50% · Zinc Nugget 25% · Uncommon Card Pack 0.25% · Common Card Pack 0.5%
Zinc Mesh	Zinc Nugget 35% · Crushed Raw Iron 10% · Gold Nugget 20% · Rare Card Pack 0.25% · Uncommon Card Pack 0.5% · Common Card Pack 1%
Brass Mesh	Crushed Raw Zinc 10% · Crushed Raw Iron 30% · Gold Nugget 50% · Emerald 1% · Rare Card Pack 0.5% · Uncommon Card Pack 1% · Common Card Pack 2.5%
Advanced Brass Mesh	Raw Zinc 40% · Raw Iron 50% · Raw Gold 30% · Emerald 5% · Diamond 1% · Ultimate Card Pack 0.1%
String Mesh waterlogged	Slimeball 10% · Clay Ball 30% · Kelp 2.5% · Water Card Pack 0.1%
Andesite Mesh waterlogged	Slimeball 20% · Clay Ball 40% · Rubber Ball 20% · Water Card Pack 0.25%
Zinc Mesh waterlogged	Slimeball 40% · Clay Ball 20% · Rubber Ball 40% · Water Card Pack 0.5%
Brass Mesh waterlogged	Slimeball 20% · Clay Ball 5% · Rubber Ball 60% · Prismarine Crystals 10% · Prismarine Shard 2.5% · Heart of the Sea 0.1% · Nautilus Shell 1% · Water Card Pack 1%

Dust block

String Mesh	Gunpowder 25% Stick 25% Common Card Pack 0.1%
Andesite Mesh	Redstone Dust 20% Gunpowder 15% Stick 15% Uncommon Card Pack 0.25% Common Card Pack 0.5%
Zinc Mesh	Glowstone Dust 20% Redstone Dust 40% Gunpowder 5% Rare Card Pack 0.25% Uncommon Card Pack 0.5% Common Card Pack 1%
Brass Mesh	Glowstone Dust 40% Redstone Dust 20% Powdered Obsidian 20% Rare Card Pack 0.5% Uncommon Card Pack 1% Common Card Pack 2.5%
Advanced Brass Mesh	Glowstone Dust 20% Redstone Dust 10% Powdered Obsidian 50% Ultimate Card Pack 0.1%

Soul Sand

Brass Mesh	Nether Quartz 10% Nugget of Experience 10% Nether Wart 5%
Advanced Brass Mesh	Nether Quartz 45% Nether Quartz 15% Ghast Tear 5% Nugget of Experience 20%

Moon Sand

Advanced Brass Mesh	Raw Iron 20% Amethyst Shard 10% Cheese 20% Desh Nugget 10% Ice Shard 10% Ultimate Card Pack 0.1%

Mars Sand

Advanced Brass Mesh	Raw Iron 20% Diamond 2.5% Ostrum Nugget 7.5% Ice Shard 20% Ultimate Card Pack 0.1%

Conglomerate

Advanced Brass Mesh	Mars Sand 20% Clay Ball 20% Ice Shard 10% Ostrum Nugget 5% Ancient Debris 0.5% Ultimate Card Pack 0.1%

Venus Sand

Advanced Brass Mesh	Coal 25% Diamond 5% Calorite Nugget 5% Raw Gold 25% Ultimate Card Pack 0.1%

Crushed end stone

Advanced Brass Mesh	Ender Pearl 10% Shulker Shell 1% Chorus Fruit 1% Ultimate Card Pack 0.3%

Appendix B · Mechanical Extruder

The Mechanical Extruder makes a block forever. The block below it is a catalyst and is not used up.

Makes	Block below (kept)	Needs next to it	Added by
Andesite	Block of Andesite Alloy	1000 mB Lava + 1000 mB Water	Quicreate
Asurine	Block of Zinc	1000 mB Lava + 1000 mB Water	Quicreate
Basalt	Soul Soil	1000 mB Lava + Blue Ice	Quicreate
Cobblestone	Cobblestone	1000 mB Lava + 1000 mB Water	Quicreate
Conglomerate	Conglomerate	1000 mB Lava + 1000 mB Water	Quicreate
Crimsite	Block of Iron	1000 mB Lava + 1000 mB Water	Quicreate
Diorite	Block of Quartz	1000 mB Lava + 1000 mB Water	Quicreate
End Stone	End Stone	1000 mB Lava + 1000 mB Water	Quicreate
Limestone	*nothing*	1000 mB Lava + 1000 mB Honey	Quicreate
Mars Cobblestone	Mars Cobblestone	1000 mB Lava + 1000 mB Water	Quicreate
Moon Cobblestone	Moon Cobblestone	1000 mB Lava + 1000 mB Water	Quicreate
Netherrack	Nether Bricks	1000 mB Lava + 1000 mB Water	Quicreate
Ochrum	Block of Gold	1000 mB Lava + 1000 mB Water	Quicreate
Scoria	*nothing*	1000 mB Water + 1000 mB Honey	Quicreate
Stone	*nothing*	1000 mB Lava + 1000 mB Water	Quicreate
Veridium	Block of Copper	1000 mB Lava + 1000 mB Water	Quicreate
Veridium	Exposed Copper	1000 mB Lava + 1000 mB Water	Quicreate
Veridium	Oxidized Copper	1000 mB Lava + 1000 mB Water	Quicreate
Veridium	Weathered Copper	1000 mB Lava + 1000 mB Water	Quicreate

Appendix C · Fishing and kelpy balls

The fishing rod's tier decides which kelpy balls you can catch. The modifier first rolls the rare extras, then the heavy, medium and light ball, each with the chance shown; if nothing hits you get the normal fishing catch.

Rod	Light ball	Medium ball	Heavy ball	Rare extras
any vanilla rod	5%	–	–	–
Brass Fishing Rod	7%	3%	–	–
Desh Fishing Rod	6%	5%	1.5%	–
Ostrum Fishing Rod	5%	6%	3%	Ancient Debris 1%, Budding Amethyst 0.5%

Light Kelpy Ball (2–4 rolls; % is per roll)

Raw Cod 1–3 **13.5%** · Raw Salmon 1–2 **9.6%** · Kelp 2–6 **13.5%** · String 1–4 **9.6%** · Bone 1–3 **9.6%**
Sweet Berries 2–5 **7.7%** · Sea Pickle 1–3 **7.7%** · Cobweb **3.8%** · Iron Nugget 2–6 **7.7%**
Gold Nugget 2–5 **5.8%** · Copper Nugget 2–6 **5.8%** · Zinc Nugget 2–6 **5.8%**

Medium Kelpy Ball (1–3 rolls; % is per roll)

Name Tag **8.1%** · Saddle **6.8%** · Turtle Egg 1–2 **6.8%** · Scute 1–2 **6.8%** · Tube Coral Block **4.1%**
Brain Coral Block **4.1%** · Bubble Coral Block **4.1%** · Fire Coral Block **4.1%** · Horn Coral Block **4.1%**
Coast Armor Trim Smithing Template **4.1%** · Tide Armor Trim Smithing Template **2.7%**
a music disc (14 kinds) **5.4%** · Pointed Dripstone 1–3 **6.8%** · Iron Ingot 1–2 **10.8%** · Gold Ingot 1–2 **6.8%**
Prismarine Shard 2–5 **8.1%** · Prismarine Crystals 2–4 **6.8%**

Heavy Kelpy Ball (1–2 rolls; % is per roll)

Totem of Undying **2.6%** · Wet Sponge 1–2 **15.4%** · Trident **7.7%** · Sniffer Egg **7.7%** · Echo Shard 1–2 **10.3%**
Disc Fragment 1–3 **10.3%** · Recovery Compass **2.6%** · Goat Horn **7.7%** · Book **7.7%** · Heart of the Sea **5.1%**
Nautilus Shell 1–2 **12.8%** · Diamond 1–2 **10.3%**

Appendix D · Card packs, cards and spawn eggs

Packs give random cards when opened. The pack-to-card lists below come from the pack-content recipe files.

Where card packs come from

Pack	Sources
Common Card Pack	sift Gravel + Andesite Mesh 0.5% sift Gravel + Brass Mesh 2.5% sift Gravel + Zinc Mesh 1% sift Sand + Andesite Mesh 0.5% sift Sand + Brass Mesh 2.5% sift Sand + String Mesh 0.1% sift Sand + Zinc Mesh 1% sift Dirt + Andesite Mesh 0.5% sift Dirt + String Mesh 0.1% sift Dirt + Zinc Mesh 1% sift Dust block + Andesite Mesh 0.5% sift Dust block + Brass Mesh 2.5% sift Dust block + String Mesh 0.1% sift Dust block + Zinc Mesh 1%
Uncommon Card Pack	sift Gravel + Andesite Mesh 0.25% sift Gravel + Brass Mesh 1% sift Gravel + Zinc Mesh 0.5% sift Sand + Andesite Mesh 0.25% sift Sand + Brass Mesh 1% sift Sand + Zinc Mesh 0.5% sift Dirt + Andesite Mesh 0.25% sift Dirt + Zinc Mesh 0.5% sift Dust block + Andesite Mesh 0.25% sift Dust block + Brass Mesh 1% sift Dust block + Zinc Mesh 0.5%
Rare Card Pack	sift Gravel + Brass Mesh 0.5% sift Gravel + Zinc Mesh 0.25% sift Sand + Brass Mesh 0.5% sift Sand + Zinc Mesh 0.25% sift Dirt + Zinc Mesh 0.25% sift Dust block + Brass Mesh 0.5% sift Dust block + Zinc Mesh 0.25%
Nether Card Pack	Blast Furnace: Uncommon Card Pack Furnace: Uncommon Card Pack
End Card Pack	**no source found**
Christmas Card Pack	Crafting Table (shapeless): Base Card Pack + White Dye + Paper + Powder Snow Bucket
Ultimate Card Pack	sift Gravel + Advanced Brass Mesh 0.1% sift Conglomerate + Advanced Brass Mesh 0.1% sift Crushed end stone + Advanced Brass Mesh 0.3% sift Dust block + Advanced Brass Mesh 0.1% sift Mars Sand + Advanced Brass Mesh 0.1% sift Moon Sand + Advanced Brass Mesh 0.1% sift Sand + Advanced Brass Mesh 0.1% sift Venus Sand + Advanced Brass Mesh 0.1%
Food Card Pack	**no source found**
Water Card Pack	sift Dirt + String Mesh (waterlogged) 0.1% sift Sand + Andesite Mesh (waterlogged) 0.25% sift Sand + Brass Mesh (waterlogged) 1% sift Sand + String Mesh (waterlogged) 0.1% sift Sand + Zinc Mesh (waterlogged) 0.5%
Space Card Pack	**no source found**
July 4th Card Pack	Crafting Table (shapeless): Base Card Pack + Gunpowder + Red Dye + White Dye + Blue Dye
Base Card Pack	**no source found**

Spawn eggs from cards

Spawn egg	Card(s) that craft into it, and the packs that contain them

Allay Spawn Egg	Allay Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box
Axolotl Spawn Egg	Blue Axolotl Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box Brown Axolotl Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box Cyan Axolotl Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box Gold Axolotl Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box Pink Axolotl Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box Rainbow Axolotl Card not in any pack list in the files
Bat Spawn Egg	Bat Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box
Bee Spawn Egg	Angry Bee Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box Bee Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box
Blaze Spawn Egg	Blaze Card Nether Card Pack, Nether Mega Box, Ultimate Card Pack, Ultimate Mega Box Blizz Card not in any pack list in the files
Cat Spawn Egg	Black Cat Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box British Cat Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box Calico Cat Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box Jelly Cat Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box Persian Cat Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box Ragdoll Cat Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box Red Cat Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box Siamese Cat Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box Tabby Cat Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box Tuxedo Cat Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box White Cat Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box
Cave Spider Spawn Egg	Cave Spider Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box
Chicken Spawn Egg	Chicken Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box
Cod Spawn Egg	Cod Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box
Cow Spawn Egg	Cow Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box
Creeper Spawn Egg	Charged Creeper Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box Christmas Creeper Card not in any pack list in the files Creeper Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box Exploding Creeper Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box Exploding Void Creeper Card End Card Pack, End Mega Box, Ultimate Card Pack, Ultimate Mega Box Golden Creeper Card not in any pack list in the files Huge Creeper Card not in any pack list in the files Huge Golden Creeper Card not in any pack list in the files Void Creeper Card End Card Pack, End Mega Box, Ultimate Card Pack, Ultimate Mega Box
Dolphin Spawn Egg	Dolphin Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box
Drowned Spawn Egg	Drowned Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box
Elder Guardian Spawn Egg	Elder Guardian Card Ultimate Card Pack, Ultimate Mega Box
Enderman Spawn Egg	Christmas Enderman Card not in any pack list in the files Enderman Card End Card Pack, End Mega Box, Ultimate Card Pack, Ultimate Mega Box
Endermite Spawn Egg	Endermite Card End Card Pack, End Mega Box, Ultimate Card Pack, Ultimate Mega Box
Fox Spawn Egg	Fox Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box Snowy Fox Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box
Frog Spawn Egg	Cold Frog Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box Rainbow Frog Card not in any pack list in the files Temperate Frog Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box Warm Frog Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box
Ghast Spawn Egg	Ghast Card Nether Card Pack, Nether Mega Box, Ultimate Card Pack, Ultimate Mega Box Ghast Shooting Card Nether Card Pack, Nether Mega Box, Ultimate Card Pack, Ultimate Mega Box
Glow Squid Spawn Egg	Glow Squid Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box
Goat Spawn Egg	Goat Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box
Guardian Spawn Egg	Guardian Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box

Husk Spawn Egg	Husk Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box
Iron Golem Spawn Egg	Christmas Iron Golem Card not in any pack list in the files Iron Golem Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box
Llama Spawn Egg	Llama Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box
Magma Cube Spawn Egg	Christmas Magma Cube Card not in any pack list in the files Magma Cube Card Nether Card Pack, Nether Mega Box, Ultimate Card Pack, Ultimate Mega Box
Mooshroom Spawn Egg	Mooshroom Card Ultimate Card Pack, Ultimate Mega Box
Phantom Spawn Egg	Phantom Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box
Pig Spawn Egg	Pig Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box
Piglin Brute Spawn Egg	Piglin Brute Card Nether Card Pack, Nether Mega Box, Ultimate Card Pack, Ultimate Mega Box
Piglin Spawn Egg	Piglin Card Nether Card Pack, Nether Mega Box, Ultimate Card Pack, Ultimate Mega Box
Pillager Spawn Egg	Pillager Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box
Polar Bear Spawn Egg	Polar Bear Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box
Pufferfish Spawn Egg	Pufferfish Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box
Rabbit Spawn Egg	Black And White Rabbit Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box Black Rabbit Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box Gold Rabbit Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box Killer Rabbit Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box Rabbit Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box Salt And Pepper Rabbit Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box
Ravager Spawn Egg	Ravager Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box
Salmon Spawn Egg	Christmas Salmon Card not in any pack list in the files Salmon Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box
Sheep Spawn Egg	Golden Sheep Card not in any pack list in the files Rainbow Sheep Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box Sheep Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box
Shulker Spawn Egg	Shulker Card End Card Pack, End Mega Box, Ultimate Card Pack, Ultimate Mega Box Shulker Shelled Card End Card Pack, End Mega Box, Ultimate Card Pack, Ultimate Mega Box
Silverfish Spawn Egg	Silverfish Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box
Skeleton Spawn Egg	Skeleton Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box
Slime Spawn Egg	Christmas Slime Card not in any pack list in the files Rainbow Slime Card not in any pack list in the files Slime Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box
Sniffer Spawn Egg	Sniffer Card Ultimate Card Pack, Ultimate Mega Box
Snow Golem Spawn Egg	Snowgolem Card not in any pack list in the files
Spider Spawn Egg	Invisible Spider Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box Spider Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box
Squid Spawn Egg	Squid Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box
Stray Spawn Egg	Stray Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box
Strider Spawn Egg	Christmas Strider Card not in any pack list in the files Golden Strider Card not in any pack list in the files Strider Card Nether Card Pack, Nether Mega Box, Ultimate Card Pack, Ultimate Mega Box
Tadpole Spawn Egg	Tadpole Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box
Turtle Spawn Egg	Turtle Card Ultimate Card Pack, Ultimate Mega Box, Water Card Pack, Water Mega Box
Vex Spawn Egg	Charging Vex Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box Vex Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box
Villager Spawn Egg	Card Trader Villager Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box Huge Villager Card not in any pack list in the files Villager Card Rare Card Pack, Rare Mega Box, Ultimate Card Pack, Ultimate Mega Box

Wither Skeleton Spawn Egg	Wither Skeleton Card Nether Card Pack, Nether Mega Box, Ultimate Card Pack, Ultimate Mega Box
Wolf Spawn Egg	Angry Wolf Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box Wolf Card Common Card Pack, Common Mega Box, Ultimate Card Pack, Ultimate Mega Box
Zombie Spawn Egg	Zombie Card Ultimate Card Pack, Ultimate Mega Box, Uncommon Card Pack, Uncommon Mega Box
Zombified Piglin Spawn Egg	Zombified Piglin Card not in any pack list in the files

Appendix E · Bosses and their drops

Chances are per kill and include every pool in the boss's loot table.

Boss	Drops (loot table)
Ender Guardian The Moon · #127 Ender Guardian	Gauntlet of Guard 1 100% · Music Disc 0–1 10%
Ender Dragon The Moon · #140 Free the End	no loot table in the files (drops handled by game code)
Ancient Remnant Mars · #151 Ancient Remnant	Sandstorm in a Bottle 1 100% · Remnant's Skull 1 100% · Block of Ancient Metal 1 100% · Music Disc 0–1 10%
The Harbinger Venus · #169 The Harbinger	Block of Witherite 1 100% · Music Disc 0–1 10%
Umvuthi, the Sunbird Venus · #174 Umvuthi	Sol Visage 1 100% · Feather 0–4 100%
Ignis Mercury · #177 Ignis	Ignitium Ingot 3 100% · Music Disc 0–1 10%
Netherite Monstrosity Mercury · #178 Netherite Monstrosity	Infernal Forge 1 100% · Monstrous Horn 1 100% · Lava Power Cell 16–24 100% · Music Disc 0–1 10%
Frostmaw Glacio · #190 Frostmaw	Ice Crystal 1 100% · Music Disc 1 100%
Ferrous Wroughtnaut Glacio · #191 Ferrous Wroughtnaut	Wrought Helm 1 100% · Axe of a Thousand Metals 1 100%
Maledictus Glacio · #193 Maledictus	Music Disc 0–1 10% · Cursium Ingot 3–4 100%
The Leviathan Glacio · #194 The Leviathan	Tidal Claws 1 100% · Abyssal Egg 1 100% · Music Disc 0–1 10%
Scylla Glacio · #195 Scylla	Lacrima 8–16 100% · Music Disc 0–1 10% · Essence of the Storm 3–4 100%
Wither · #218 The Wither	no loot table in the files (drops handled by game code)

Appendix F · Items with no source, and progression notes

Items that a quest or a shown recipe needs, but that have no recipe, ore, mob drop, loot or other source in the files the builder read.

Every quest item and ingredient traced in this guide has at least one recipe or natural source in the pack files.

Hard-to-get items checked

Common skyblock bottlenecks, with every source found in the pack files.

Item	Sources found in the files
Nether Wart	Sift Soul Sand with Brass Mesh: **5%** Nether island: 2× Nether Wart
Sponge	Furnace: Wet Sponge
Wet Sponge	Open a Heavy Kelpy Ball (fishing): 22.2% per ball Elder Guardian drops it (100% per kill; where it spawns is not in the data files)
Honeycomb	Bees (from a Bee card → Bee Spawn Egg) fill a beehive or nest; shear a full one. (game mechanic)
Totem of Undying	Open a Heavy Kelpy Ball (fishing): 3.8% per ball Evoker drops it (100% per kill; where it spawns is not in the data files)
Lava Bucket	Starting island: 1× in the starting chest Fill a bucket with Lava (Spout, or pick it up)
Blaze Rod	Blaze drops it (100% per kill; where it spawns is not in the data files)
Obsidian	Mechanical Press + Basin (compacting) (Superheated): 4× Powdered Obsidian
Slimeball	Sift Sand with Zinc Mesh (waterlogged): **40%** Sift Sand with Andesite Mesh (waterlogged): **20%** Sift Sand with Brass Mesh (waterlogged): **20%** Crafting Table (shapeless): Dough + Lime Dye Slime drops it (75% per kill; where it spawns is not in the data files)
Leather	Millstone: Saddle → 2 Chest loot in the Abandoned Spire (Glacio), 54% per chest Chest loot in the Abandoned Spire (Glacio), 48.3% per chest Chest loot in the Abandoned Temple (Glacio), 54% per chest Chest loot in the Abandoned Temple (Glacio), 48.3% per chest Chest loot in the Abandoned Village (Glacio), 54% per chest
Wither Skeleton Skull	Wither Skeleton (Mercury) drops it, 2.5% per kill
Nether Star	Dropped by the Wither when you kill it (handled by the game code, see **#218 The Wither**). (game mechanic)
Ender Pearl	Sift Crushed end stone with Advanced Brass Mesh: **10%** Enderman (The Moon) drops it, 100% per kill
Glowstone Dust	Sift Dust block with Brass Mesh: **40%** Sift Dust block with Advanced Brass Mesh: **20%** Sift Dust block with Zinc Mesh: **20%** Witch drops it (23.4% per kill; where it spawns is not in the data files)
Diamond	Sift Venus Sand with Advanced Brass Mesh: **5%** Sift Mars Sand with Advanced Brass Mesh: **2.5%** Sift Sand with Advanced Brass Mesh: **1%** Open a Heavy Kelpy Ball (fishing): 15% per ball Crushing Wheels: Diamond Horse Armor Mars: Mars Diamond Ore → Diamond (Y -80 to 80)
Emerald	Sift Sand with Advanced Brass Mesh: **5%** Sift Sand with Brass Mesh: **1%** Crafting Table (shapeless): Block of Emerald → 9 Chest loot in the Lunar Tower (The Moon), 60.3% per chest Chest loot in the Lunar Tower (The Moon), 20.9% per chest Chest loot in the Lunarian Village (The Moon), 60.3% per chest
Lapis Lazuli	Sift Gravel with Advanced Brass Mesh: **10%** Crafting Table (shapeless): Block of Lapis Lazuli → 9 Glacio: Glacio Lapis Ore → Lapis Lazuli (Y -32 to 32) Glacio: Deepslate Lapis Lazuli Ore → Lapis Lazuli (Y -80 to 10) Chest loot in the Moon Dungeon (The Moon), 11.9% per chest Chest loot in the Sunken City (Glacio), 63.5% per chest

Ancient Debris	Sift Conglomerate with Advanced Brass Mesh: **0.5%** Fishing with the Ostrum Fishing Rod: 1% per catch Mars: Ancient Debris (Y 90 to 130)
Platinum	Furnace: Raw Platinum
Desh Ingot	Furnace: Raw Desh Chest loot in the Lunar Tower (The Moon), 86.7% per chest Chest loot in the Lunar Tower (The Moon), 20.9% per chest Chest loot in the Lunarian Village (The Moon), 86.7% per chest Chest loot in the Lunarian Village (The Moon), 20.9% per chest
Ostrum Ingot	Furnace: Raw Ostrum
Calorite Ingot	Furnace: Raw Calorite
Bone Meal	Millstone: Koboleton Bone → 4 Lionfish drops it (5% per kill; where it spawns is not in the data files) Raw Cod drops it (5% per kill; where it spawns is not in the data files) Pufferfish drops it (5% per kill; where it spawns is not in the data files) Raw Salmon drops it (5% per kill; where it spawns is not in the data files) Tropical Fish drops it (5% per kill; where it spawns is not in the data files)
Sugar Cane	Sift Sand with String Mesh: **2.5%**
Prismarine Shard	Sift Sand with Brass Mesh (waterlogged): **2.5%** Open a Medium Kelpy Ball (fishing): 15.6% per ball Elder Guardian drops it (100% per kill; where it spawns is not in the data files) Guardian drops it (100% per kill; where it spawns is not in the data files)
Shulker Shell	Sift Crushed end stone with Advanced Brass Mesh: **1%** Shulker drops it (50% per kill; where it spawns is not in the data files)

Progression notes

- **Umvuthi.** The quest subtitle says "Optional boss in the Venus wastelands", but the datapack's `mowziesmobs/worldgen/structure/umvuthana_grove.json` only allows the grove in `quicreate:venus_fungal_forest`. (`config/mowziesmobs-common.toml` still whitelists `ad_astra:venus_wastelands` for Umvuthi.)
- **Card packs on a Deployer.** Quicreate replaces each pack-content recipe in `data/cards/recipes/` with a Deployer + Shears recipe on the pack. All cards of one pack share the same two inputs, so the files do not say which card a deployer will give.
- **Quests without arrows.** Several 1.2 quests have no dependencies in the quest book (Electricity, Crushing Wheels, Mechanical Crafters, Carbon Brush, Wires, Mechanical Extruder). The guide places them after the quests that make their ingredients.
- **Planet mobs without drops.** Ad Astra's mobs (Pygro, Mogler, Martian Raptor, Glacian Ram, Corrupted Lunarian…) have no entity loot tables in any jar, so the guide lists them with no drops.

Appendix G · Build notes

Files this guide was built from

Quest book: `config/heracles` (220 quests). CraftTweaker: `scripts/` (246 scripts: 72 removals, 174 added recipes). Mods: 63 jars in `mods/`. Quicreate datapack: jar:quicreate-1.7.jar (969 data files), quicreate-src (91 data files). Vanilla data: vanilla:mcmeta_data, vanilla:mcmeta_assets, vanilla:resources.

Jake's PrismLauncher libraries folder (with the vanilla client jar and Forge) was not shared with this build, so vanilla 1.20.1 recipes, loot tables, tags, models, textures and names were taken from the misode/mcmeta 1.20.1 data and assets branches (extracted from the official client jar), and Forge's own item tags (forge:ingots/iron and friends) from the MinecraftForge 1.20.1 branch's generated resources. On Jake's PC the builder finds the real client jar and Forge universal jar in the libraries folder by itself.

Load order (later wins for recipes, loot tables and worldgen; tags merge): vanilla, then mod jars alphabetically, then the Quicreate jar (only with `--quicreate-jar-fallback`), then the Quicreate source folder.

Quicreate cleanup script

`quicreate_cleanup_2026-11-09.ps1` lists 69 recipe files to rename, 115 data files and 13 scripts to delete. The guide treats every listed file (and the old name of every renamed file) as already removed. At build time none of those files existed any more, so the cleanup appears to have been applied already.

Recipes removed by CraftTweaker

Calorite Engine (ad_astra:calorite_engine), Calorite Tank (ad_astra:calorite_tank), Etrionic Combustion Generator (ad_astra:coal_generator), Compressor (ad_astra:compressor), Cryo Freezer (ad_astra:cryo_freezer), Desh Cable (ad_astra:desh_cable), Desh Engine (ad_astra:desh_engine), Desh Tank (ad_astra:desh_tank), Energizer (ad_astra:energizer), Etrionic Blast Furnace (ad_astra:etrionic_blast_furnace), Etrionic Capacitor (ad_astra:etrionic_capacitor), Steel Rocket Propeller (ad_astra:fan), Fuel Refinery (ad_astra:fuel_refinery), Gravity Normalizer (ad_astra:gravity_normalizer), Jet Suit Boots (ad_astra:jet_suit_boots), Jet Suit Pants (ad_astra:jet_suit_pants), Rocket Assembly Bench (ad_astra:nasa_workbench), Ostrum Engine (ad_astra:ostrum_engine), Ostrum Tank (ad_astra:ostrum_tank), Oxygen Distributor (ad_astra:oxygen_distributor), Oxygen Loader (ad_astra:oxygen_loader), Oxygen Sensor (ad_astra:oxygen_sensor), Photovoltaic Etrium Cell (ad_astra:photovoltaic_etrium_cell), Rocket Fin (ad_astra:rocket_fin), Rocket Nose Cone (ad_astra:rocket_nose_cone), Solar Panel (ad_astra:solar_panel), Space Helmet (ad_astra:space_helmet), Steel Cable (ad_astra:steel_cable), Steel Engine (ad_astra:steel_engine), Steel Tank (ad_astra:steel_tank), Water Pump (ad_astra:water_pump), Wheel (ad_astra:wheel), Base Card Pack (cards:base_card_pack_recipe), Common Card Pack (cards:comon_mob_card_packs_recipe), End Card Pack (cards:end_mob_card_packs_recipe), Food Card Pack (cards:food_card_pack), Nether Card Pack (cards:nether_mob_card_packs_recipe), Rare Card Pack (cards:rare_mob_card_packs_recipe), Ultimate Card Pack (cards:ultimate_card_packs_recipe), Uncommon Card Pack (cards:uncommon_mob_card_packs_recipe), Water Card Pack (cards:water_card_pack), Eye of Abyss (cataclysm:abyss_eye), Eye of Void (cataclysm:void_eye), Mechanical Extruder (create_mechanical_extruder:crafting/mechanical_extruder), Advanced Energiser (create_new_age:shaped/advanced_energiser), Advanced Solar Heating Plate (create_new_age:shaped/advanced_solar_heating_plate), Basic Motor (create_new_age:shaped/basic_motor), Basic Motor Extension (create_new_age:shaped/basic_motor_extension), Basic Solar Heating Plate (create_new_age:shaped/basic_solar_heating_plate), Carbon Brushes (create_new_age:shaped/carbon_brushes), Electrical Connector (create_new_age:shaped/electrical_connector), Electrical Connector (create_new_age:shaped/electrical_connector_mirrored), Fluxuated Magnetite (create_new_age:shaped/fluxuated_magnetite), Generator Coil (create_new_age:shaped/generator_coil), Heat Pipe (create_new_age:shaped/heat_pipe), Heat Pipe (create_new_age:shaped/heat_pipe_mirrored), Heat Pump (create_new_age:shaped/heat_pump), Heater (create_new_age:shaped/heater), Layered Magnet (create_new_age:shaped/layered_magnet), Netherite Magnet (create_new_age:shaped/netherite_magnet), Reactor Fuel Acceptor (create_new_age:shaped/reactor_fuel_acceptor), Redstone Magnet (create_new_age:shaped/redstone_magnet), Reinforced Energiser (create_new_age:shaped/reinforced_energiser), Stirling Engine (create_new_age:shaped/stirling_engine), Reactor Heat Vent (create_new_age:shapeless/reactor_heat_vent), Advanced Brass Mesh (createsifter:advanced_brass_mesh), Andesite Mesh (createsifter:andesite_mesh), Brass Mesh (createsifter:brass_mesh), Brass Sifter (createsifter:brass_sifter), String Mesh (createsifter:string_mesh), Zinc Mesh (createsifter:zinc_mesh), Netherite Upgrade Smithing Template (minecraft:netherite_upgrade_smithing_template)

Limits

The builder only knows what is in data files. Things done by game code are not in this book unless a quest or this guide's mechanics list mentions them: villager trades, the odds of opening card packs, the Ender Dragon and Wither drops, bucket filling, axe stripping, mobs that structures spawn from code, and spawn-egg colours (spawn eggs are drawn untinted). Drop chances are worked out from the loot tables without looting or fortune.